

Volume 52, issue 10
15 June 2008

ISSN 0167-9473

COMPUTATIONAL STATISTICS & DATA ANALYSIS

Incorporating Statistical Software Newsletter

Contains Special Section:

Special Section on Microsoft Excel 2007
Guest Editor: B.D. McCullough

The official journal of



The International Association for Statistical Computing
A Section of The International Statistical Institute

Available online at

 **ScienceDirect**
www.sciencedirect.com

Special Section:
Special Section on Microsoft Excel 2007

Guest Editor:
B.D. McCullough



Editorial

Special section on Microsoft Excel 2007

"[I]t is quite possible that more basic statistical calculations are done worldwide in EXCEL than in all statistical packages combined". – Wilkinson (1994)

"Let's not kid ourselves: the most widely used piece of software for statistics is Excel". Ripley (2002)

At least since Sawitzki's (1994a; 1994b) project, which was published in the pages of this journal, it has been known that serious errors exist in Excel's statistical procedures. As has been recounted in these pages since then, Microsoft has not been very good at fixing the errors, often not bothering to fix known errors and sometimes introducing new errors while attempting to fix old ones. Consequently, when a new version of Excel is released, everyone wants to know, "Has Microsoft fixed the errors?" Papers and notes on various aspects of Excel's statistical functionality might appear scattered in various outlets over time. For the release of Excel 2007, the special section has sought to collect the various papers and publish them together.

David Heiser, author of the website, "Microsoft Excel 2000, 2003 and 2007 Faults, Problems, Workarounds and Fixes" (Heiser, 2006) and McCullough have written the first paper (McCullough and Heiser, 2008), in which they assess some statistical procedures in Excel 2007, reprising McCullough and Wilson (2005). However, they also report errors in several functions that McCullough and Wilson never considered. For example, the "normal probability plot" is customarily used to compare regression residuals to a normal distribution; in Excel, it compares the dependent variable to a uniform distribution.

In the second paper, Yalta (2008), who has published on accuracy in other journals (e.g., Yalta (2007) and Yalta and Yalta (2007)), reviews the checkered history of Excel's statistical distributions and examines Excel 2007. Microsoft has on more than one occasion "fixed" a problem by replacing one inaccurate algorithm with a different inaccurate algorithm, and often fails to make any fixes at all. He finds that Microsoft still has not fixed all the errors in the Excel statistical distributions.

In Excel 2003, Microsoft attempted to implement the Wichmann-Hill random number generator (WH-RNG). Instead of producing numbers between zero and unity, it would also produce negative numbers. Microsoft soon issued a patch; but how can users tell that the patch is correct? Did Microsoft just trap the negative numbers or did it correctly implement the WH-RNG? This problem is tackled in the third paper by McCullough (2008). He finds that Microsoft has again failed to implement correctly the Wichmann-Hill RNG.

Graphics packages for creating advertisements or pretty patterns have their own purposes. Graphics for displaying data fall under the discipline of statistical graphics, which has its own membership organization (ASA Section on Statistical Graphics), books, and journal (*Journal of Computational and Graphical Statistics*). One would hope that Microsoft would avail itself of this learned material, but Microsoft has resisted the temptation.

Stephen Few, a consultant who specializes in data visualization for business and who has written books on the subject recently wrote of a 2005 correspondence he conducted about Excel 2007 graphics (Few, 2006):

I learned who at Microsoft was responsible for the development of the new charting engine and quickly sent him an e-mail. In it, I introduced myself and offered to send him my suggestions for shaping Excel's new charting functionality. With his polite assent, I submitted a list of recommendations. In response, I received a very courteous e-mail. Here's an excerpt:

Thanks. This was quite interesting and useful. I think you'll be pretty happy with some of the changes we are making for Office 12. I did read your book for inspiration (as well as Tufte, Cleveland, Wilkinson, Zelazny, etc).

So Microsoft is aware of this body of knowledge. In the fourth paper, Su (2008), who studied statistical graphics under Andrew Gelman at Columbia University, reports that Microsoft has chosen to ignore the precepts of statistical graphics. He examines the extent to which Excel's graphical defaults facilitate or hinder the production of good graphics, and concludes

that the defaults lend themselves to chartjunk. It is possible to make good graphs by invoking the appropriate options, but to invoke these appropriate options the user would have to be well-versed in statistical graphics.

The fifth paper, on teaching statistics with Excel, is written by Nash (2008). Nash is a professor at the University of Ottawa, where he has taught statistics, forecasting and managing technological risk since 1980. His books, articles and papers cover many topics, but statistical computing is a common theme. Having built and used computers since the 1960s, he took an early interest in spreadsheets, and has published several articles on the uses of spreadsheets. He often uses spreadsheets in his courses. Currently he is working on the open-source "TelTable" project to provide an audit trail for spreadsheets, as well as another project for testing spreadsheets.

Over the years, this journal has published many articles describing the errors in Microsoft Excel, errors that raise concerns about Microsoft's quality assurance procedures, at least insofar as Excel is concerned. Hence it is perhaps worth comparing quality assurance in Microsoft's Excel to quality assurance in its game division. A recent issue of *Wired* magazine (Thompson, 2007) describes the way that "Bungie", a Microsoft gaming subsidiary, tests one of its products, a game called Halo3:

Because it [Bungie] is owned by Microsoft, which launches dozens Xbox and PC games every year, Bungie has access to one of the most advanced game-testing facilities ever built. [Bungie has] now analyzed more than 3000 hours of Halo3 played by some 600 everyday gamers, tracking everything from favored weapons to how and where – down to the square foot – players most frequently get killed.

Bungie doesn't just test its own games this way. It also buys copies of rival titles and studies those, too, to see how Halo3 matches up.

It is difficult not to think that if Microsoft tested business software the way it tested game software, then the statistical functions in Excel would be as accurate as those found in any other major software package. If that were the case, then none of the articles in this special section would have been written.

References

- Few, Stephen, (2006). Excel's new charting engine preview of an opportunity missed, Business Intelligence Network, UK edition, 9 May 2006. <http://www.b-eye-network.co.uk/view-articles/2818> (accessed 30.08.07).
- Heiser, David A., 2006. Microsoft excel 2000 and 2003 faults, problems, workarounds and fixes. *Computational Statistics and Data Analysis* 51 (2), 1442–1443.
- McCullough, B.D., 2008. Microsoft excel's 'not the Wichmann-Hill' random number generators. *Computational Statistics and Data Analysis* 52 (10), 4587–4593.
- McCullough, B.D., Heiser, David A., 2008. On the accuracy of statistical procedures in microsoft excel 2007. *Computational Statistics and Data Analysis* 52 (10), 4570–4578.
- McCullough, B.D., Wilson, Berry, 2005. On the accuracy of statistical procedures in microsoft excel 2003. *Computational Statistics and Data Analysis* 49 (4), 1244–1252.
- Nash, John, 2008. Teaching statistics with excel 2007 and other spreadsheets. *Computational Statistics and Data Analysis* 52 (10), 4602–4606.
- Ripley, Brian D., (2002). Statistical methods need software: A view of statistical computing. Opening lecture RSS 2002, Plymouth.
- Sawitzki, Gunther, 1994a. Testing numerical reliability of data analysis systems. *Computational Statistics and Data Analysis* 18 (2), 269–286.
- Sawitzki, Gunther, 1994b. Report on the reliability of data analysis systems. *Computational Statistics and Data Analysis* 18 (2), 289–301.
- Su, Yu-Sung, 2008. It's easy to produce chartjunk using microsoft excel 2007 but hard to make good graphs. *Computational Statistics and Data Analysis* 52 (10), 4594–4601.
- Thompson, Clive, 2007. The science of play. *Wired* 15 (9), 140. et seq.
- Wilkinson, Leland, 1994. Practical guidelines for testing statistical software. In: Dirschedel, P., Ostermann, R. (Eds.), *Computational Statistics*. Berlin, Physica-Verlag.
- Yalta, A.Talha, 2007. The numerical reliability of GAUSS 8.0. *The American Statistician* 61 (3), 262–268.
- Yalta, A.Talha, Yalta, A.Yasemin, 2007. GRETL 1.6.0 and its numerical accuracy. *Journal of Applied Econometrics* 22 (4), 849–854.
- Yalta, A.Talha, 2008. The accuracy of statistical distributions in microsoft excel 2007. *Computational Statistics and Data Analysis* 52 (10), 4579–4586.

B.D. McCullough*
LeBow College of Business,
Department of Decision Sciences,
Drexel University,
Philadelphia, PA, 19104-2875,
United States
E-mail address: bdmccullough@drexel.edu.
Available online 12 March 2008

* Tel.: +1 215 895 2134.



On the accuracy of statistical procedures in Microsoft Excel 2007

B.D. McCullough^{a,*}, David A. Heiser^b

^a Department of Decision Sciences, LeBow College of Business, Drexel University, Philadelphia, PA 19104, United States

^b Carmichael, CA, United States

ARTICLE INFO

Article history:
Available online 12 March 2008

ABSTRACT

Excel 2007, like its predecessors, fails a standard set of intermediate-level accuracy tests in three areas: statistical distributions, random number generation, and estimation. Additional errors in specific Excel procedures are discussed. Microsoft's continuing inability to correctly fix errors is discussed. No statistical procedure in Excel should be used until Microsoft documents that the procedure is correct; it is not safe to assume that Microsoft Excel's statistical procedures give the correct answer. Persons who wish to conduct statistical analyses should use some other package.

© 2008 Elsevier B.V. All rights reserved.

1. Introduction

McCullough (1998) proposed an intermediate-level set of tests for assessing statistical software. This methodology has been applied to statistical and econometric software (McCullough, 1999a,b; Kitchen et al., 2003; Keeling and Pavur, 2007; Yalta, 2007). The methodology assesses the software in three areas: statistical distributions, random number generation, and the NIST StRD ("Statistical Reference Datasets", www.itl.nist.gov/div898/strd). Typically, when errors are found in a software package, the developer fixes the errors, and the matter is done; there is no further need to re-evaluate the software on this set of tests. Not so with Microsoft and its product Excel. Microsoft occasionally fixes errors, more often ignores them, and sometimes fixes them incorrectly. Consequently, every time there is a new version of Excel, the tests must be repeated.

Indeed, every time a new version of Excel is released, we receive emails asking, "Is it safe to use Excel?" There appears to be a sentiment that if only Excel would pass the intermediate tests, then it would be safe to use. Nothing could be farther from the truth. Microsoft has not even fixed all the flaws identified by Sawitzki (1994) over 15 years ago (his paper took a couple years to be published). There are scores of statistical functions in Excel, and we have benchmarked only a fraction of them. In the fraction that we have examined, we have found enough unfixed/incorrectly-fixed errors to cast grave doubt on the above-mentioned sentiment. To wit, we write this paper in part to warn members of the statistical community that even should there come a day when Microsoft can fix enough of the errors in Excel that Excel can pass these intermediate-level tests, it will not necessarily be safe to use.

In this paper we repeat the intermediate level tests. We also examine some other statistical procedures in which Excel has long been deficient. This is by no means intended to be a comprehensive list but, rather, merely a sampling of errors that we have encountered and that Microsoft has not bothered to fix. Given Microsoft's track record, we recommend that no statistical procedure be used unless Microsoft demonstrates that the procedure in question has been correctly programmed, e.g., by the use of test datasets from textbooks, or by comparison with another software package.

In Section 2 we discuss the intermediate level tests and examine a major flaw in the Excel Solver. In Section 3 we show that Excel's procedure for performing Exponential Smoothing is grievously flawed; we wonder how such obvious errors could have been made. Section 4 similarly analyzes Excel's flawed LOGEST function. Section 5 discusses Excel's "Normal

Probability Plot" ATP function, not from a graphical perspective but from a statistical perspective. In the statistical literature, the usual Normal Probability Plot compares regression residuals to a normal distribution as a way to check the normality of the residuals. Excel's version compares the dependent variable to a uniform distribution, though we are unaware of any reason that a user might wish to check the dependent variable for uniformity. In Section 6 we recount some miscellaneous errors that are by no means exhaustive, but merely illustrative of the difficulties that await the unsuspecting Excel user. Section 7 presents the conclusions.

2. Intermediate level tests

In this section, we update previous assessments of Excel (McCullough and Wilson, 1999, 2002, 2005), and elaborate on Excel Solver's inability to identify that it has stopped at a point that is not a solution.

Excel's statistical distributions have always been inadequate. Over the years, Microsoft has fixed some distributions, fixed others incorrectly, and failed to fix others. For further details, see the accompanying article by Yalta (2008). The performance of Excel 2007 on this suite of tests can be judged inadequate.

The random number generator has always been inadequate. With Excel 2003, Microsoft attempted to implement the Wichmann–Hill generator and failed to implement it correctly. The "fixed" version appears in Excel 2007 but this "fix" was done incorrectly. Microsoft has twice failed to implement correctly the dozen lines of code that constitute the Wichmann–Hill generator; this is something that any undergraduate computer science major should be able to do. The Excel random number generator does not fulfill the basic requirements for a random number generator to be used for scientific purposes: (1) it is not known to pass standard randomness tests, e.g., L'Ecuyer and Simard's (2007) CRUSH tests (these supersede Marsaglia's (1996) DIEHARD tests—see Altman et al. (2004) for a comparison); (2) it is not known to produce numbers that are approximately independent in a moderate number of dimensions; (3) it has an unknown period length; and (4) it is not reproducible. For further discussion of these points, see the accompanying article by McCullough (2008); the performance of Excel 2007 in this area is inadequate.

The NIST StRD has five suites of test problems: univariate summary statistics, one-way ANOVA, linear regression, nonlinear least squares, and Markov-chain Monte Carlo (this last suite is not applicable here). Each of these suites contains several test problems with known solutions. The performance of Excel 97, Excel 2000 and Excel XP in all four areas was unacceptable. In Excel 2003, Microsoft made improvements to the univariate, ANOVA and linear regression procedures and performed satisfactorily in those areas. However, with default settings Solver returns zero digits of accuracy for over half of the 27 nonlinear test problems and, even tuned for better accuracy, still returns zero digits of accuracy for 11 of the 27 problems (Berger, 2007). Consequently, Excel 2003 failed the StRD portion of the tests.

Applying the StRD to Excel 2007 indicates that Microsoft made no changes in this area for Excel 2007. The performance of Excel 2007 in this area is inadequate. It has been noted that Excel Solver has a marked tendency to stop at a point that is not a solution and declare that it has found a solution. See also Mondragon and Borchers (2005). Many books recommend Solver for solving nonlinear statistical problems; see, *inter alia*, Barreto and Howland (2006), de Levie (2003), and Gottfried (2002). While these books tend to be used in applied settings (i.e., not for publication in scholarly journals), Solver is used in academic journal articles: for solving a GARCH problem in finance (Jones et al., 1998); for factor analysis (Miles, 2005); for modeling kinetic chemical reactions (Denton, 2000); in chromatography (Nikitas and Pappa-Louisi, 2000). It is not difficult to find uses of Solver for statistical purposes. Here we briefly consider the "Misra1a" StRD nonlinear problem that is discussed at length in McCullough (2004a):

$$y = \beta_1(1 - \exp(-\beta_2 x)) + \epsilon \quad (1)$$

for $n = 14$ observations and starting values of 500 for β_1 and 0.0001 for β_2 . The correct solution is $\hat{\beta}_1 = 238.94212918$, $\hat{\beta}_2 = 0.00055015643181$ and sum of squared residuals = 0.12455138894.

The Solver default algorithm is a "Newton" method, with an option for a "Conjugate" method. The default convergence tolerance ("ct") is 0.001. Running from default, Solver stops in eight iterations, yielding a "solution" $\hat{\beta}_1 = 454.12$ and $\hat{\beta}_2 = 0.0002676$; Solver specifically states: "Solver has converged to the current solution. All constraints are satisfied." Such a message might lead a researcher to conclude that the software had found a solution to the nonlinear least squares problem Misra1a, but such a conclusion would be incorrect. If the researcher could examine the gradient (which should componentwise be zero at a point that minimizes the sum of squared residuals), he would find that the gradient was nowhere zero, not even close to zero. However, Solver does not allow the user to inspect the gradient.

If the researcher simply changed the algorithm to "Conjugate" then Solver would stop in seven iterations, and report "Solver has converged to the current solution. All constraints and optimality conditions are satisfied." [italics added] This seemingly more impressive declaration of having found a solution would be even more misleading than the prior declaration. This time Solver reports $\hat{\beta}_1 = 500$ and $\hat{\beta}_2 = 0.0002422$. Apparently the first "solution" did not satisfy some "optimality conditions" but this second solution does, despite the fact that Solver has not budged from the starting point for β_1 , and despite the fact that the sum of squared residuals has increased from 16.725 to 19.516. Of course, the gradient at this point is not even close to zero.

Nonlinear solvers may stop for a variety of reasons other than having reached a solution (see McCullough and Renfro (2000, Section 7) for a discussion), and a reliable solver can distinguish whether a stopping point is or is not a solution.

* Corresponding author. Tel.: +1 215 895 2134.

E-mail addresses: bdmccullough@drexel.edu (B.D. McCullough), dah_box1@innercite.com (D.A. Heiser).

As mentioned above, Solver does find the correct solution for 16 of the 27 nonlinear problems. However, the only reason we are able to know that Solver has actually found the correct solution is because we know what the correct solutions are in advance. If we had to rely on Solver's convergence messages, we would have no confidence that it had stopped at the correct solution. Therefore, let us be explicit: do not use the Excel Solver to solve nonlinear least squares problems. Additionally, in the absence of evidence that Solver can reliably solve a particular class of nonlinear problems, we would hesitate to use it for any such problem.

3. Exponential smoothing

Exponential Smoothing is a classical statistical procedure. Here we follow the standard derivation given by Montgomery et al. (1990, p. 82). For a series X_t , model it as

$$X_t = b + \epsilon_t \quad (2)$$

where b is the expected demand in any period and ϵ_t is an error term. In period t , we have the estimate of b made in the previous period, $\hat{b}_{t-1}$, as well as the current value of the series, X_t . The forecast error is given by $e_t = X_t - \hat{b}_{t-1}$. Using this information to create an updated forecast of the current period, $\hat{b}_t$, we can modify the old forecast by some proportion α of the forecast error to yield

$$\hat{b}_t = \hat{b}_{t-1} + \alpha(X_t - \hat{b}_{t-1}). \quad (3)$$

Setting $\hat{b}_t = S_t$ produces the more usual equation

$$S_t = S_{t-1} + \alpha(X_t - S_{t-1}) \quad (4)$$

or

$$S_t = \alpha X_t + (1 - \alpha)S_{t-1}, \quad 0 \leq \alpha \leq 1 \quad (5)$$

where α is commonly referred to a "smoothing constant". Obviously the initial value of S_t must be set in some fashion, and there are many such ways discussed in the literature. For our purposes it suffices to note that simply starting with $t = 1$ and setting $X_1 = S_1$ and then calculating $S_2, S_3, \dots$ is perfectly acceptable (Newbold and Bos, 1990, p. 161).

Microsoft's implementation of Exponential Smoothing is found in the Analysis ToolPak, and is worth quoting extensively from the documentation; see Fig. 1.

All this seems well and good. Let us now use Excel to perform Exponential Smoothing on the thirty observations of sales data given by Newbold and Bos (1990, Table 6.1), presented in our Table 1. Newbold and Bos also provide the smoothed values for $\alpha = 0.3$, and these values were the same ones produced by MINITAB, RATS, and every other statistical package we tried; these also are given in Table 1.

Upon selecting the "Exponential Smoothing" option of the ATP, Excel asks for an "input range", a "damping factor" and an "output range". It seems reasonable to suppose that the "damping factor" is the "smoothing constant" referred to in the Excel help files, so we entered 0.3. The first thing to notice about the Excel output is that it does not give a smoothed value for the first observation, and gives the smoothed value of 86 as 103. If the procedure is initialized in the usual fashion, setting $X_1 = S_1$, then we would expect 103 to be the smoothed value for the first observation. While this is definitely incorrect, we might try to make sense of this by deducing that Microsoft's mistake was to shift all the smoothed values by one place. However, the smoothed value after 103 is not the same as that given by Newbold and Bos (and the other statistical packages). A little reverse engineering reveals that the relationship between the "damping factor" that the program asks for, and the "smoothing constant" referred to in the documentation is

$$\text{damping factor} = 1 - \text{smoothing constant}. \quad (6)$$

Indeed, when we use a damping factor of 0.7, we get the correct numbers, but they are still in the incorrect positions. Employing equation (6) is blatantly misleading, and even if one is warned about this, the Excel output is still wrong: it gives incorrect smoothed values for all the observations.

The Exponential Smoothing analysis tool predicts a value based on the forecast for the prior period, adjusted for the error in that prior forecast. The tool uses the smoothing constant α , the magnitude of which determines how strongly forecasts respond to errors in the prior forecast. Note Values of 0.2 to 0.3 are reasonable smoothing constants. These values indicate that the current forecast should be adjusted 20 to 30 percent for error in the prior forecast. Larger constants yield a faster response but can produce erratic projections. Smaller constants can result in long lags for forecast values.

Fig. 1. Help file for exponential smoothing.

Table 1
Exponential smoothing results

Obs	Sales	Smoothed value	Excel	Excel
		Newbold/Bos	Damping = 0.3	Damping = 0.7
1	103	103	N/A	N/A
2	85	97.90	103.00	103.00
3	92	93.73	91.10	97.90
4	84	90.62	95.13	93.73
5	92	91.17	84.64	90.62
6	95	91.17	84.64	90.62
26	116	108.30	105.13	105.00
27	100	105.82	102.74	108.30
28	128	112.47	103.82	105.82
29	109	111.43	120.75	112.47
30	122	114.60	112.62	111.43

We searched several textbooks, and could not find any exponential smoothing example that even remotely resembles what Microsoft has programmed into Excel for this function. Indeed, we cannot imagine that there is a textbook which gives Excel's output for this particular input. We are left to wonder, How does Microsoft check the accuracy of the procedures it programs into Excel? Does anyone use the program before writing the help files? What else might account for the "damping factor" / "smoothing constant" confusion?

4. LOGEST

Excel's LOGEST and GROWTH functions are bad; here we only discuss the LOGEST function, which has been analyzed separately by Cook et al. (1999) and Hesse (2006) and who reached the same conclusion. To see why the LOGEST function is bad, it is useful to start with the Excel Help file, given in Fig. 2.

LOGEST: In regression analysis, calculates an exponential curve that fits your data and returns an array of values that describes the curve. Because this function returns an array of values, it must be entered as an array formula.

The equation for the curve is:

$$y = b * m^x \quad \text{or} \quad y = (b * (m1^x1) * (m2^x2) * \dots) \quad (\text{if there are multiple x-values})$$

where the dependent y-value is a function of the independent x-values. The m-values are bases corresponding to each exponent x-value, and b is a constant value. Note that y, x, and m can be vectors. The array that LOGEST returns is mn,mn-1,...,m1,b.

Fig. 2. Excel help file for LOGEST.

As is obvious, the equation $y = b * m^x$ is a nonlinear equation and needs to be solved nonlinearly. We know that the Excel Solver is not reliable and its results would have to be checked by using a reliable package, so let us solve the problem in a reliable package to begin with. We choose "R", and use the example data from the Excel Help files.

```
units <- c(33100,47300,69000,102000,150000,220000)
month <- c(11,12,13,14,15,16)
summary(nls(units~b*m^month,start=list(b=500,m=1.5))
```

Formula: units ~ b * m^month

Parameters:

```
Estimate Std. Error t value Pr(>|t|)
b 473.71781 10.86801 43.59 1.66e-06 ***
m 1.46784 0.00221 664.09 3.08e-11 ***
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 503.2 on 4 degrees of freedom

However, the additional statistics LOGEST returns are based on the following linear model:
 $\ln y = x_1 \ln m_1 + \dots + x_n \ln m_n + \ln b$
 You should keep this in mind when you evaluate the additional statistics, especially the *se* and *seb* values, which should be compared to $\ln m_i$ and $\ln b$, not to m_i and b . For more information, consult an advanced statistics manual.

Fig. 3. Excel help file on LOGEST standard errors.

In sharp contrast, the Excel LOGEST output is

	estimate	std error
b	495.3048	0.035834
m	1.463276	0.002633

The coefficients are markedly different. We can indulge a presumption that when "R" and Excel disagree, Excel is wrong. The sum of squared residuals from the estimated models yields 2,758,373 for Excel and 1,012,875 for R; our presumption appears to be justified. Consequently, it appears that Excel coefficients are accurate to one digit for *b* and three digits for *m*. Abstracting from the issue of how to calculate standard errors for the coefficients in a nonlinear model, and simply assuming that R is correct and Excel is wrong, we see that the Excel standard errors are not accurate even to one digit! Why do these differences occur?

The reason for the differences in the coefficients and sum of squared errors is that Excel uses a crude linear approximation. LOGEST actually runs a linear regression on

$$\log(\text{units}) = c + n \text{ month} \quad (7)$$

to obtain $c = 6.2052$ (0.035834) and $n = 0.3807$ (0.002633) [standard errors in parentheses] and then uses exponentiation to obtain $b = \exp(c) = 495.3048$ and $m = \exp(n) = 1.463276$. However, this crude approximation does not minimize the sum of squared residuals. The Excel approach minimizes

$$\sum (\log_i y_i - (x_i \log m + \log b))^2$$

which is clearly not the same thing as minimizing

$$\sum (y_i - b m^{x_i})^2$$

though, from reading the Excel Help files, Microsoft clearly believes that the two methods are equivalent.

As to the differences in the standard errors, the relevant information for *m* in the Microsoft Help File is given in Fig. 3.

Of course, if the relevant persons at Microsoft had themselves consulted an "advanced statistics manual" they would have discovered their error. Nonetheless, Excel is presenting the standard errors from the regression given by Eq. (7) with the exponentiated coefficients from that regression. To present standard errors that are not to be interpreted on the same basis as the coefficients is highly unusual, even misleading, given the customary statistical practice of dividing a coefficient by its standard error to compute a *t*-statistic. This information about the presented standard errors is hidden at the bottom of the Help file and a user who is accustomed to standard statistical output might be forgiven for calculating *t*-statistics as the ratio of the presented coefficient to the presented standard error. Further, Microsoft makes the point that it exponentiates the coefficients to obtain the desired estimates; a novice might well reason by analogy that he should exponentiate the presented standard errors. Of course, exponentiating the standard errors is not the right thing to do, but let us do it anyhow: for *b* we have $\exp(0.035834) = 1.036484$ and for *m* we have $\exp(0.002633) = 1.002636$. Why does Microsoft even bother to present these useless and misleading standard errors? We are unable to find or imagine any reason for such a practice.

The GROWTH function suffers from the same methodological flaw: using a crude and incorrect approximation instead of the correct approach (Cook et al., 1999).

5. Normal probability plot

We discuss this here because from our perspective, this is not a graphical issue but a statistical issue: what exactly are the statistics that underlie the Normal Probability Plot? After running a regression, it is often useful to check the residuals for normality. A common way to do this is to graph a Normal Probability Plot of the residuals. The Normal Probability Plot puts the expected value of the order statistic of the standard normal distribution on the abscissa with the corresponding data value on the ordinate. If the data are normally distributed, then the plot should be a straight line. Departures from a straight line indicate departures from normality.

In the ATP Regression procedure there is a box for "Normal Probability Plots". The entire Help File for this function, given in Fig. 4, is especially useless.

Normal Probability Plots

Select to generate a chart that plots normal probability.

Fig. 4. Excel help file for normal probability plot.

In a departure from customary practice, the user is not told how the plot is generated, nor even of what variable is plotted. But it would not be outlandish for a trusting user to suppose that he would see the usual normal probability plot of the residuals. Such trust would be betrayed.

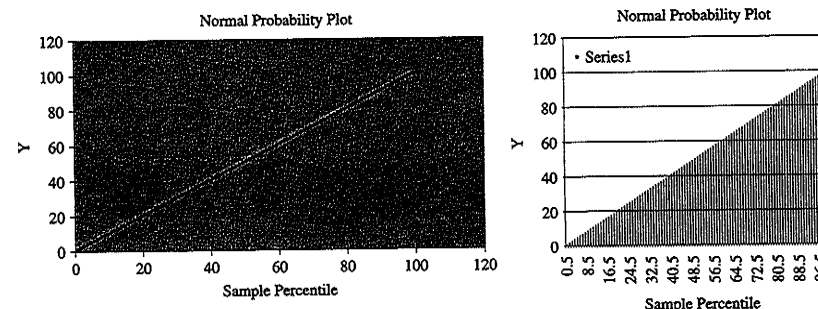


Fig. 5. Excel 2003 (left) and Excel 2007 (right) "normal probability plots" for sequence 1–100.

To investigate this feature, we ran a regression whose dependent variable is the integers 1 through 100 and whose independent variable is 100 random uniforms. The Normal Probability Plot from this regression has "Sample Percentile" on the abscissa and "Y" on the ordinate. Note that the confusing chart Excel 2003 (see Fig. 5(left)) is even more confusing in the "improved" Excel 2007 (see Fig. 5(right)): why the vertical bars for something that should be represented by a line? The trusting user might think that perhaps this is a Normal Probability Plot of the dependent variable, but he would be wrong. It is somewhat difficult to see given the poor aspect ratio, but the plot is a straight line. The Normal Probability plot is only supposed to yield a straight line if the input variable is normally distributed, and in this instance the input variable is the sequence 1 through 100. Some insight into this mystery can be gained from the "Probability Output" table that corresponds to Fig. 5 is given in Table 2.

Table 2
"Probability output" from Excel ATP regression

Percentile	Y
0.5	1
1.5	2
2.5	3
3.5	4
4.5	5
5.5	6
6.5	7
7.5	8
8.5	9
9.5	10
10.5	11
11.5	12
12.5	13
13.5	14
14.5	15
15.5	16
16.5	17
17.5	18
18.5	19
19.5	20
20.5	21
21.5	22
22.5	23
23.5	24
24.5	25
25.5	26
26.5	27
27.5	28
28.5	29
29.5	30
30.5	31
31.5	32
32.5	33
33.5	34
34.5	35
35.5	36
36.5	37
37.5	38
38.5	39
39.5	40
40.5	41
41.5	42
42.5	43
43.5	44
44.5	45
45.5	46
46.5	47
47.5	48
48.5	49
49.5	50
50.5	51
51.5	52
52.5	53
53.5	54
54.5	55
55.5	56
56.5	57
57.5	58
58.5	59
59.5	60
60.5	61
61.5	62
62.5	63
63.5	64
64.5	65
65.5	66
66.5	67
67.5	68
68.5	69
69.5	70
70.5	71
71.5	72
72.5	73
73.5	74
74.5	75
75.5	76
76.5	77
77.5	78
78.5	79
79.5	80
80.5	81
81.5	82
82.5	83
83.5	84
84.5	85
85.5	86
86.5	87
87.5	88
88.5	89
89.5	90
90.5	91
91.5	92
92.5	93
93.5	94
94.5	95
95.5	96
96.5	97
97.5	98
98.5	99
99.5	100

Obviously the data in the "Percentile" column are not the expected values of the order statistics of the standard normal, and neither are they the percentiles corresponding to the expected values of the order statistics of the standard normal. What, then, are they?

A bit of math shows that the "Percentile" column is governed by the formula

$$\text{Percentile} = \frac{100i - 50}{n} \quad \forall i = 1, 2, \dots, n.$$

In truth, the plot should be labeled "Uniform Probability Plot" because Microsoft is comparing the dependent variable to a uniform distribution. In sum, as far as the statistical theory of the Excel "Normal Probability Plot" is concerned, Excel computes the plot for the wrong variable, the dependent variable instead of the residuals, and has managed to confuse the

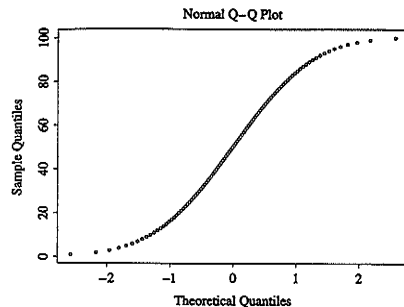


Fig. 6. Correct "normal probability plot" for sequence 1–100.

uniform distribution for the normal distribution. For purposes of comparison, the correct normal probability plot for the sequence 1 through 100 is given by R, and is presented in Fig. 6. The nonnormality of the data is evident, whereas Excel incorrectly indicated that the uniform data were normal.

6. Miscellaneous

Here we present several other errors in Excel of which we are aware. Again, we stress that this is not a comprehensive list. We simply wish to warn the reader that errors in Excel are not limited to those we have discussed above, and that a particular procedure is not necessarily safe to use just because no errors have been reported in newsgroups or elsewhere. We note that Microsoft does not make available to Excel users a list of the errors of which Microsoft is aware. The most comprehensive such resource of which we are aware is Heiser's (2006a) website, "Microsoft Excel 2000, 2003 and 2007 Faults, Problems, Workarounds and Fixes," (www.daheiser.info/excel/frontpage.html).

- **Bad random normal generator in ATP.** Use the pull-down menus for "Tools", "Data Analysis", and "Random Number Generation"; enter "10" for number of variables, "2000" for number of random variables, leave mean set to zero and standard deviation set to unity, enter "123" for "Random Seed" and click ok. Inspect the resulting worksheet. Cells 253:H, 1212:G, and 1245:D will show "−9.53674" and cell 1845:H will show "5.364418". Values this extreme should not be seen in a properly functioning standard normal random number generator. McCullough and Wilson (2002) noted this problem years ago in Excel 2000/XP, and Microsoft has neither fixed it nor noted in the help files that this function is bad.
- Heiser (2006b) notes several problems with two-sample test procedures in Excel 2003, including misleading documentation and inaccurate routines, all of which are present in Excel 2007. Below is a partial list:
 - **Inaccurate t-test results in the presence of missing values.** Does not correctly calculate t test values when there are missing data cells in the range. Microsoft identified this fault (KBA 829252) in previous Excel versions, but did not fix it.
 - **Inaccurate p-values from a t-test.** Excel uses the Welch test method, which calculates a non-integer degrees-of-freedom value. The Excel algorithm uses the TTEST function, which truncates the degrees-of-freedom input to an integer. This gives an incorrect p value for the test.
 - **Incorrect labeling of t-test and z-test tables.** For the t-test, Excel describes the one tail test as " $P(T \leq t)$ one-tail" and incorrectly describes the two tail test as " $P(T \leq t)$ two-tail".
- **Bad regression results from ATP "Trendline" function for ill-conditioned data.** Excel graphics have a feature whereby right-clicking on a data point enables the user to fit one of six trendlines to the data: Linear, Logarithmic, Polynomial, Power, Exponential, and Moving Average, as an option the fitted equation and its R^2 can be placed on the figure. Pottel (2003), using an ill-conditioned dataset from Simonoff (2002), shows that the Linear trendline calculates a different equation than either the ATP Regression procedure or the LINEST command. For the Simonoff data, Trendline calculates

$$\hat{y} = 1E + 09 - 0.125x \quad R^2 = -0.6667$$

(the negative sign in the R^2 is not a typo) while the other two methods yield

$$\hat{y} = 9903477999 + 0.096522x \quad R^2 = 0.000283.$$

- **Bad Exponential and Power results from ATP "Trendline" function even for datasets that are not ill-conditioned.** There are five Trendlines that ATP can place on a graph. Hesse (2006) shows that the Exponential Trendline and Power Trendline are incorrect. Does any user of Excel wish to bet that the remaining three Trendlines are correctly implemented?
- **Bad linear regression algorithm.** Until version Excel 2003, Excel could not pass the StRD linear regression problem "Filip", which is a 10th order polynomial that is nearly singular. Excel would give completely incorrect coefficients. Microsoft improved the linear regression algorithm in Excel 2003, and both Excel 2003 and Excel 2007 correctly solve the Filip

problem. None of the StRD problems tests for perfectly collinear data. A reliable linear regression routine should be able to detect perfectly collinear data (for which a traditional regression solution does not exist) and refuse to report a solution. Such a test is Test IV-C from Wilkinson's (1985) Tests (see also McCullough (2004b)). Excel 4.0 failed this test ((Sawitzki, 1994), and so does Excel 2007. Excel reports an answer, which is necessarily incorrect. If Excel warned the user, "rank deficiency encountered" then it would be okay. On the basis that the linear regression routine cannot recognize a singular regression, we can judge the linear regression routine in Excel to be unacceptable.¹

- **Bad linear regression output.** When Excel detects a singular design matrix, it removes a column from the model and sets that coefficient and standard error to zero, as per KB 828533. In this situation, the ATP fails to subtract one from the numerator degrees of freedom and, as a consequence, the "ANOVA OUTPUT" (for testing the significance of the regression) is incorrect: the regression sum of squares is correct, but its mean-square and consequently the F-statistic are incorrect, too, as is the p-value for the F-statistic.
- **Non-standard operator precedence.** In FORTRAN and most computer languages with which we are familiar, if $X = -1$ then X^2 evaluates to +1 and $-X^2$ evaluates to −1. In Excel, $-X^2$ evaluates to +1. See Berger (2007) for further discussion. While Microsoft is free to use whatever rules it choose for operator precedence, to the extent that they are nonstandard, users should be clearly warned.
- **Bad default options for graphics.** Representing data graphically is more than just flashy presentation. In fact, it is a well-defined discipline called "statistical graphics" that has its own organization (American Statistical Association Section on Statistical Graphics), its own journal (*Journal of Computational and Statistical Graphics*), and several books. Canned graphical procedures in a software package should make it easy for users to produce graphics that embody the best practices of the discipline of statistical graphics. Specifically, the software, and not the user, should be "knowledgeable" about how to display data well. Microsoft has chosen a contrary route; only users who already are intimately familiar with the discipline of statistical graphics can produce good graphs in Excel, and then only after the tedious exercise of figuring out the appropriate way to override the Excel defaults. See the article by Su (2008) in this issue.

7. Conclusions

The statistical literature has regularly identified flaws in Excel's statistical procedures at least since Sawitzki (1994), and Microsoft has repeatedly proved itself incapable of providing reliable statistical functionality. It is little wonder that introductory texts on statistics warn students not to use Excel when the results matter (e.g., Keller (2001) and Levine et al. (2002)). There is nothing inherently difficult about programming a spreadsheet that precludes reliable statistical programming. The open-source spreadsheet package Gnumeric was originally such a good Excel clone that it even reproduced many Excel errors. When informed that these were errors, the handful of part-time programmers who maintain Gnumeric managed to fix them correctly in a matter of months (McCullough, 2004c). What is particularly pernicious, at least so far as consumers of statistics are concerned, is that Microsoft seems to take the approach that specialized domain knowledge is not necessary to the production of statistical code, i.e., that the entire field of statistical computing has no merit. As but one case to illustrate this point, in KB 829215, Microsoft discusses the enhancements made to the ATP ANOVA function in Excel 2003. The KB describes pseudocode for the computations in previous versions of Excel; this pseudocode embodies an algorithm that would be implemented by no one who has read an introductory text on statistical computing, e.g., Kennedy and Gentle (1980) or Thisted (1988). Microsoft then remarks,

Again, for model 2 and model 3, sums of squares are calculated and a quantity is subtracted from the sum of squares as in the calculator formula. Unfortunately, basic statistics texts frequently suggest approaches for ANOVA such as the one that is shown earlier in this article.

The obvious implication is that Microsoft programmed its statistical functions by using statistics texts, rather than statistical computing texts. Of course, this approach to programming – eschewing specialized domain knowledge – explains much of Microsoft's inability to program functions correctly, or fix them correctly.

The improvements in Excel 2003 gave hope that Microsoft might finally be committed to providing reliable statistical functionality in Excel. That Microsoft once again decided to allow errors to remain unfixed, or fixed incorrectly, in the Excel 2007 release dashes those hopes. Regardless of the reasons for Microsoft's inability to program Excel correctly, Microsoft has had years to fix these errors, and has proven itself unable to do so. It is clear that when Microsoft claims to have "fixed" a procedure that is known to be in error, Microsoft's claim cannot be safely believed. It is also clear that there are procedures in Excel that are in error, and Microsoft either does not know of the error or does not choose to warn users that the procedure is in error. Since it is generally impossible for the average user to distinguish between the accurate and inaccurate Excel functions, the only safe course for Excel users is to rely on no statistical procedure in Excel unless Microsoft provides tangible evidence of its accuracy consisting of:

¹ How exactly Excel should inform the user of a singular regression is not clear. Should the program generate a specific error message instead of producing any output, or should some of the cells in the output return the "#NUM" message? Either approach would bring a program to a crashing halt, but perhaps this would be a good thing. Usually, singular data imply that whoever entered the data made some sort of mistake and, if the input data are bad, the program probably should not run to completion.

1. test data with known input and output;
2. description of the algorithm sufficient to permit a third party to use the test data to reproduce Excel's output; and
3. a bona fide reference for the algorithm.

If Microsoft does not perform these actions for each statistical procedure in Excel, then there are only two safe alternatives for the user who is concerned about the accuracy of his statistical results: the user can perform all these actions himself, or simply not use Excel.

Acknowledgements

Thanks for comments to the referee, and also to Patrick Burns, Jerry Lewis, John Nash, Yu-Sung Su, and A. Talha Yalta.

References

- Altman, Micah, Gill, Jeff, McDonald, Michael P., 2004. Numerical Issues in Statistical Computing for the Social Scientists. John Wiley and Sons, New York.
- Barreto, Humberto, Howland, Frank M., 2006. Introductory Econometrics: Using Monte Carlo Simulation with Microsoft Excel. Cambridge University Press, New York.
- Berger, Roger L., 2007. Nonstandard operator precedence in Excel. *Computational Statistics & Data Analysis* 51 (6), 2788–2791.
- Cook, H.R., Cox, M.G., Dainton, M.P., Harris, P.M., 1999. Testing spreadsheets and other packages used in metrology: Testing the intrinsic functions of Excel. Report to the National Measurement System Policy Unit, Department of Trade and Industry, from the UK Software Support for Metrology Programme. NPL Report CISE 27/99.
- de Levie, Robert, 2003. Advanced Excel for Scientific Analysis. Oxford University Press, New York.
- Denton, Philip, 2000. Analysis of first-order kinetics using Microsoft Excel solver. *Journal of Chemical Education* 77 (11), 1524–1525.
- Gottfried, Byron S., 2002. Spreadsheet Tools for Engineers Using Excel. McGraw-Hill, New York.
- Heiser, David A., 2006a. Microsoft Excel 2000 and 2003 faults, problems, workarounds and fixes. *Computational Statistics & Data Analysis* 51 (2), 1442–1443.
- Heiser, David A., 2006b. Statistical tests, tests of significance, and tests of a hypothesis using Excel. *Journal of Applied Statistical Methods* 5 (2), 155–171.
- Hesse, Rick, 2006. Incorrect nonlinear trend curves in Excel. *Foresight: International Journal of Applied Forecasting* 3, 39–43.
- Jones, Charles M., Lamont, Owen, Lumsdaine, Robin, 1998. Macroeconomic news and bond market volatility. *Journal of Financial Economics* 47, 315–337.
- Keeling, Kellie, Pavur, Robert, 2007. A comparative study of the reliability of nine statistical software packages. *Computational Statistics & Data Analysis* 51 (8), 3811–3831.
- Keller, G., 2001. Applied Statistics with Microsoft Excel. Duxbury, Belmont, CA.
- Kennedy, William, Gentle, James, 1980. Statistical Computing. Marcel Dekker, New York.
- Kitchen, A., Drachenberg, R., Symanzik, J., 2003. Assessing the reliability of web-based statistical software. *Computational Statistics* 18 (1), 107–122.
- L'Ecuier, Pierre, Simard, Richard, 2007. TESTU01: A C library for empirical testing of random number generators. *ACM TOMS* 33 (4), Article 22.
- Levine, D., Stephan, D., Krehbiel, T., Berenson, M., 2002. Statistics for Managers Using Microsoft Excel, 3e. Prentice-Hall, Englewood Cliffs, NJ.
- Marsaglia, George, 1996. DIEHARD: A Battery of Tests of Randomness. <http://stat.fsu.edu/pub/diehard>.
- McCullough, B.D., 1998. Assessing the reliability of statistical software: Part I. *The American Statistician* 52, 358–366.
- McCullough, B.D., 1999a. Assessing the reliability of statistical software: Part II. *The American Statistician* 53 (2), 149–159.
- McCullough, B.D., 1999b. Econometric software reliability: E-Views, LIMDEP, SHAZAM, and TSP. *Journal of Applied Econometrics* 14 (2), 191–202.
- McCullough, B.D., 2004a. Some details of nonlinear estimation, chapter eight. In: Altman, Micah, Gill, Jeff, McDonald, Michael P. (Eds.), *Numerical Methods in Statistical Computing for the Social Sciences*. Wiley, New York.
- McCullough, B.D., 2004b. Wilkinson's tests and econometric software. *Journal of Economic and Social Measurement* 29 (1–3), 261–270.
- McCullough, B.D., 2004c. Fixing statistical errors in spreadsheet software: The cases of gnumeric and Excel. *CSDA Statistical Software Newsletter*, www.csdassn.org/software_reports.cfm.
- McCullough, B.D., 2008. Microsoft Excel's "Not The Wichmann–Hill" random number generators. *Computational Statistics & Data Analysis*.
- McCullough, B.D., Renfro, Charles G., 2000. Some numerical aspects of nonlinear estimation. *Journal of Economic and Social Measurement* 26 (1), 63–77.
- McCullough, B.D., Wilson, Berry, 1999. On the accuracy of statistical procedures in Microsoft Excel 97. *Computational Statistics and Data Analysis* 31, 27–37.
- McCullough, B.D., Wilson, Berry, 2002. On the accuracy of statistical procedures in Microsoft Excel 2000 and XP. *Computational Statistics and Data Analysis* 40 (4), 27–37.
- McCullough, B.D., Wilson, Berry, 2005. On the accuracy of statistical procedures in Microsoft Excel 2003. *Computational Statistics and Data Analysis* 49 (4), 1244–1252.
- Miles, J.N.V., 2005. Confirmatory factor analysis using Microsoft Excel. *Behavior and Research Methods* 37 (4), 672–676.
- Mondragon, P.F., Borchers, B., 2005. A comparison of nonlinear regression codes. *Journal of Modern Applied Statistical Methods* 4 (1), 343–351.
- Montgomery, Douglas C., Johnson, Lynwood A., Gardiner, John S., 1990. *Forecasting and Time Series Analysis*, 2nd ed. McGraw-Hill, New York.
- Newbold, Paul, Bos, Theodore, 1990. *Introductory Business Forecasting*. Southwestern, Cincinnati, OH.
- Nikitas, P., Pappa-Louisi, A., 2000. Non-linear least-squares fitting with Microsoft Excel solver and related routines in HPLC modelling of retention. *Chromatographia* 52 (7–8), 477–486.
- Pottel, Hans, 2003. Statistical flaws in Excel, Mimeo, Innogenetics NV, Zwijnaarde, Belgium.
- Sawitzki, Gunther, 1994. Report on the numerical reliability of data analysis systems. *Computational Statistics and Data Analysis* 18 (2), 289–301.
- Simonoff, Jeffrey, 2002. Statistical analysis using Microsoft Excel 2000. Mimeo, Stern School of Business, NYU.
- Su, Yu-Sung, 2008. It's easy to produce chartjunk using Microsoft Excel 2007 but hard to make good graphs. *Computational Statistics and Data Analysis*.
- Thisted, R.A., 1988. *Elements of Statistical Computing: Numerical Computation*. Chapman and Hall, New York.
- Wilkinson, Leland, 1985. *Statistics Quiz*. Systat, Inc., Evanston, IL.
- Yalta, A. Talha, 2007. The numerical reliability of GAUSS 8.0. *The American Statistician* 61 (3), 262–268.
- Yalta, A. Talha, 2008. On the accuracy of statistical distributions in Microsoft Excel 2007. *Computational Statistics & Data Analysis*.



Contents lists available at ScienceDirect
Computational Statistics and Data Analysis
 journal homepage: www.elsevier.com/locate/csda



The accuracy of statistical distributions in Microsoft® Excel 2007

A. Talha Yalta

TOBB University of Economics and Technology, Sogutozu Caddesi No: 43, Sogutozu, 06560, Ankara, Turkey

ARTICLE INFO

Article history:
 Available online 12 March 2008

ABSTRACT

We provide an assessment of the statistical distributions in Microsoft® Excel versions 97 through 2007 along with two competing spreadsheet programs, namely Gnumeric 1.7.11 and OpenOffice.org Calc 2.3.0. We find that the accuracy of various statistical functions in Excel 2007 range from unacceptably bad to acceptable but significantly inferior in comparison to alternative implementations. In particular, for the binomial, Poisson, inverse standard normal, inverse beta, inverse student's *t*, and inverse *F* distributions, it is possible to obtain results with zero accurate digits as shown with numerical examples.

© 2008 Elsevier B.V. All rights reserved.

1. Introduction

Considered as a major revision, the new Excel 12, released along with the Microsoft Office 2007 suite, promises a large number of new features as well as improved functionality. The Excel 2007 product guide¹ mentions, among other things, an overhauled user interface, better data integration, faster calculation performance, and “dramatic visual effects” without giving a word's consideration to numerical accuracy. Given the fact that Excel is the ubiquitous application for managing and analyzing data in the world of education, business, and the government, it is highly probable that more statistical calculations are done using Excel than all statistical software combined. Therefore, the main objective of this study is to provide an assessment of the statistical distributions in Excel versions 97 through 2007 and discuss Microsoft's performance on correcting reported flaws effectively and in a timely fashion. Software testing is important as it reflects the ongoing concern of the academic community regarding the reliability of commonly used programs providing statistical functionality.

In the following section, we report on the earlier studies by various authors regarding errors in Excel's statistical distributions. In section three, we examine the reliability of various statistical distributions in Excel versions 97, 2000, 2002, 2003, and 2007 along with two competing spreadsheet programs, namely Gnumeric 1.7.11 and OpenOffice.org Calc 2.3.0. Our comparison is merely illustrative and the specific distributions chosen are those highlighted by earlier studies regarding Excel's statistical functions. Section four offers the conclusions.

2. Fixing errors in Excel's statistical distributions

Keeling and Pavur (2007) and Yalta (2007) show that, in general, software vendors do respond to software testing by correcting documented flaws in their programs, resulting in improved software reliability. One exception to this is Microsoft Corporation, which, to this day, largely remained indifferent to numerous studies pointing out the unacceptable accuracy errors of the statistical procedures in their Excel product. Microsoft's cavalier attitude toward accuracy is so shocking that it prompted McCullough (2001) to ask the rhetorical question, “Does Microsoft Fix Errors in Excel?” His conclusion was that it could be a profit maximizing strategy on behalf of Microsoft to satisfy a large class of users' demand for software which is flashy and easy-to-use, if not very accurate and reliable.

¹ E-mail addresses: yalta@etu.edu.tr, tahayalta@gmail.com.

² Available for download at <http://office.microsoft.com/en-us/excel/HA101680001033.aspx> (retrieved July 4, 2007).

Table 1
Microsoft's performance on correcting errors in Excel's statistical distributions

Distribution	Excel 97	Excel 2000	Excel 2002	Excel 2003	Excel 2007
Binomial	Flaws reported	Not fixed	Not fixed	Poor fix	Not fixed
Hypergeometric	Flaws reported	Not fixed	Not fixed	Poor fix	Not fixed
Poisson	Flaws reported	Not fixed	Not fixed	Poor fix	Not fixed
Normal	Flaws reported	Not fixed	Not fixed	Fixed	Not fixed
Inv. normal	Flaws reported	Not fixed	Poor fix	Poor fix	Not fixed
Inv. chi-square	Flaws reported	Not fixed	Poor fix	Poor fix	Not fixed
Inv. t	Flaws reported	Not fixed	Not fixed	Poor fix	Not fixed
Inv. F	Flaws reported	Not fixed	Not fixed	Poor fix	Not fixed
Gamma				Flaws reported	Not fixed
Inv. beta				Flaws reported	Not fixed

The earliest study regarding Excel's statistical distributions is by Knüsel (1998), who used SAS for Windows 6.12 and his own ELV utility as a benchmark to assess the reliability of the elementary statistical distributions in Excel 97. Reporting a long list of errors in discrete distributions namely Poisson, binomial, and hypergeometric as well as continuous distributions such as standard normal, inverse standard normal, inverse chi-square, inverse F, and inverse t, Knüsel concluded, "So one has to warn statisticians against using Excel functions for scientific purposes." In a follow-up study, McCullough (2001) noted that all the errors cataloged previously remained uncorrected in Excel 2000. McCullough also found that the only observed change in Excel 2002 (included in Microsoft Office XP) was to the inverse normal distribution, which, even after being "fixed," could not return more than three digits of accuracy in the tail of the distribution. Shortly thereafter, Knüsel (2002) affirmed these results and warned once again that Excel should not be used for statistical purposes.

Wondering whether there was something inherently difficult about correcting errors in spreadsheet software, McCullough (2004) investigated how an alternative package, Gnumeric (v1.0.4), coped with the flaws surprisingly similar to those found in Excel (97/2000/2002). He found that the few part-time volunteers who maintain and develop Gnumeric were able to fix all the problems in statistical procedures within several weeks after the errors have been reported.

With the release of Office 2003, Microsoft announced extensive improvements² in Excel's statistical functions – including functions for discrete distributions, continuous distribution inverse functions, as well as the normal distribution and its related functions – that "stretch such functions to their limits." Attempting to verify Microsoft's claims, Knüsel (2005) found that some of the inaccurate results cataloged previously have been corrected while some have stayed, and some have been replaced by new errors. He also showed that the gamma and beta distributions were not always computed correctly in Excel 2003. Knüsel's results were shortly confirmed by McCullough and Wilson (2005), who advised persons desiring to conduct statistical analysis of data against using Excel 2003.

Table 1 shows Microsoft's performance on correcting published errors in Excel's statistical functions over the last five major revisions. As it will be shown in the next section, many reliability issues previously reported for different versions of Excel remain unresolved in Excel 2007. Barring the standard normal distribution, which now appears to be accurate, the accuracy of these functions range from unacceptably bad to acceptable but significantly inferior in comparison to alternative implementations.

3. Numerical accuracy of various statistical functions in Excel 2007

Knüsel (1995) asks the question "What accuracy is required of a program computing statistical distributions?" His answer is that if the computed probability is given as 0.0000 and the exact value is smaller than 0.5×10^{-4} , then the answer given by the computer is correct and reliable. On the other hand, if the program returns 4.820×10^{-15} and the correct value is 4.073×10^{-15} , then this answer is unacceptable since it makes the user believe that the result is accurate to four significant digits while in fact only one of these digits is correct. Accordingly, if the answer cannot be calculated accurately, the program should note this rather than misleading the user by returning an incorrect result.

We certainly agree with Knüsel that the answer given by a program should always be correct as it is printed out. In addition, we second McCullough and Vinod (1999) that software should be judged inadequate if a poor or a defective algorithm is used when there is a known alternative that can provide an accurate answer to a particular problem. We should like to note that today there exists well-known, freely available, and easy to find algorithms that can provide extremely accurate answers for the computation of most statistical distributions. In fact, according to today's computing standards, a good program offering statistical functionality is expected to provide six or seven digits of accuracy on commodity hardware for probabilities as small as 10^{-300} and even smaller. But how do the statistical procedures in Excel 2007 fare against today's computing standards?

Below, we provide an assessment of the accuracy of various statistical distributions in Excel versions 97 through 2007. The specific functions chosen are those previously highlighted in different studies including McCullough (2001, 2004), McCullough and Wilson (1999, 2002, 2005) and Knüsel (1998, 2002, 2005). The parameter combinations presented in the

² See Microsoft Knowledgebase Article #828888.

Table 2
Binomial distribution with parameters ($k, n = 1030, p = 0.5, \Sigma = 1$)

	EXACT	ELV Ed 2	EXCEL 97/2K/XP	EXCEL 2003/2007	CALC 2.3.0	GNUMERIC 1.7.11
1	8.96134E-308	0	8.95245E-308	0	Exact	Exact
2	4.61499E-305	0	Exact	0	Exact	Exact
100	1.39413E-169	0	Exact	0	Exact	Exact
200	5.45731E-92	Exact	Exact	0	Exact	Exact
300	2.91621E-42	Exact	Exact	0	Exact	Exact
390	3.18196E-15	Exact	Exact	0	Exact	Exact
391	5.24099E-15	Exact	Exact	2.05902E-15	Exact	Exact
400	3.89735E-13	Exact	Exact	3.86553E-13	Exact	Exact
410	3.19438E-11	Exact	Exact	3.19406E-11	Exact	Exact
420	1.76037E-09	Exact	Exact	Exact	Exact	Exact
500	1.83106E-01	Exact	#NUM!	Exact	Exact	Exact
550	9.86550E-01	Exact	#NUM!	Exact	Exact	Exact
575	9.99920E-01	Exact	#NUM!	Exact	Exact	Exact
589	0.99998E-01	Exact	#NUM!	Exact	Exact	Exact

tables are for illustrative purposes and largely follow those used by the aforementioned studies. The "exact" numbers were calculated using Mathematica 5.2, which can use symbolic methods to perform computations with arbitrary precision.³ These values were also checked with the program ELV (Knüsel, 2003), which can provide exact results with six digits of accuracy for upper and lower tail probabilities as well as upper and lower quantiles of nine elementary statistical distributions for probabilities as small as 10^{-100} .

Two additional programs included in this study are Gnumeric 1.7.11 and OpenOffice.org Calc 2.3.0, which, although being open source software, employ dissimilar subroutines for the calculation of various statistical distributions. More specifically, released under the GNU General Public License (GPL), Gnumeric is allowed to use the math/stats library of the GNU-R statistical environment, which is also a GPL program. On the other hand, because OpenOffice.org is released under the GNU Lesser General Public License (LGPL), it cannot share the libraries that are ordinary GPL licensed and therefore has to rely on its custom developed procedures for statistical distributions.⁴ Both of these alternative spreadsheet applications provide a range of features roughly equivalent to those offered by Excel.

All computations were done using an Intel Centrino Duo 2.16 GHz notebook computer with 2 GB memory and running under the GNU/Linux platform (kernel 2.6.18.8). The Windows XP operating system used for running different versions of Excel as well as the ELV utility was virtualized using VirtualBox 1.4.0.

3.1. Binomial distribution

If a random variable X has a binomial $Bi(n, p)$ distribution with the parameters n for the number of trials and p for the probability for a success, then the Excel function BINOMDIST, with the logical argument *cumulative* = TRUE ($\Sigma = 1$), computes for a given k , the probability $Pr\{X \leq k\}$. As Table 2 shows, Excel 97, 2000, and XP can correctly compute probabilities as small as 4.6×10^{-308} while giving no result for central probabilities.⁵ We also see that this bad algorithm was later replaced by an even worse algorithm, so that Excel 2003 and 2007 can mislead the users by providing results with zero accurate digits. Gnumeric and Calc, on the other hand, can provide exact results even for extremely small probability values.

3.2. Hypergeometric distribution

If a random variable X has a hypergeometric $H(N, M, n)$ distribution with the parameters n for the number of draws without replacement from a finite population of size N with M distinctive members, then the Excel function HYPGEOMDIST computes, for a given k number of successes, the point probability $Pr\{X = k\}$. For the sake of comparison with Knüsel (1998), here we consider the case where exactly k white balls are selected after making 500 draws from an urn of 1030 balls and the number of white balls in the urn is given as 515. As Table 3 shows, Excel 97, 2000, and XP cannot compute these probabilities for $N \geq 1030$ while more recent versions of Excel can return an accurate answer only for the range $187 < k < 313$. Gnumeric and Calc, on the other hand, can provide exact results for all possible values of $k \in \{0, 1, \dots, 500\}$. In addition, Gnumeric also offers the Boolean option *cumulative* for computing the tail probabilities $Pr\{X \leq k\}$, lack of which in Excel was criticized twice by Knüsel (1998, 2005) and is still missing in Excel 2007.

³ See McCullough (2000) for a detailed assessment of this program in terms of numerical accuracy.

⁴ See de Laat (2005) for a discussion of the significance of different property regimes for software development.

⁵ For more information, see Microsoft Knowledgebase Article #828888, which also provides links to pseudo-code for the algorithms used to avoid the overflow problems affecting various discrete probability distributions including the binomial, hypergeometric, and Poisson distributions in Excel 2002 and earlier.

Table 3

Hypergeometric distribution with parameters ($k, N = 1030, M = 515, n = 500$)

	EXACT	ELV Ed.2	EXCEL 97/2K/XP	EXCEL 2003/2007	CALC 2.3.0	GNUMERIC 1.7.11
0	1.60137E-280	0	#NUM!	0	Exact	Exact
100	7.46483E-83	Exact	#NUM!	0	Exact	Exact
187	1.53541E-15	Exact	#NUM!	0	Exact	Exact
188	4.13038E-15	Exact	#NUM!	Exact	Exact	Exact
200	1.65570E-10	Exact	#NUM!	Exact	Exact	Exact
300	1.65570E-10	Exact	#NUM!	Exact	Exact	Exact
312	4.13038E-15	Exact	#NUM!	Exact	Exact	Exact
313	1.53541E-15	Exact	#NUM!	0	Exact	Exact
400	7.46483E-83	Exact	#NUM!	0	Exact	Exact
500	1.60137E-280	0	#NUM!	0	Exact	Exact

Table 4

Poisson distribution with parameters (k, λ, σ)

			EXACT	ELV Ed.2	EXCEL 97/2K/XP	EXCEL 2003/2007	CALC 2.3.0	GNUMERIC 1.7.11
0	200	0	1.38390E-87	Exact	Exact	0	Exact	Exact
102	200	0	1.41720E-14	Exact	Exact	0	Exact	Exact
104	200	0	2.72538E-14	Exact	Exact	Exact	Exact	Exact
133	200	0	1.07322E-07	Exact	Exact	Exact	Exact	Exact
134	200	0	1.51227E-07	Exact	#NUM!	Exact	Exact	Exact
200	200	0	2.81977E-02	Exact	#NUM!	Exact	Exact	Exact
314	200	0	2.23568E-14	Exact	#NUM!	Exact	Exact	Exact
315	200	0	1.41948E-14	Exact	#NUM!	0	Exact	Exact
400	200	0	5.58069E-36	Exact	#NUM!	0	Exact	Exact
900	200	0	1.73230E-286	0	#NUM!	0	Exact	Exact
1E+03	1E+03	-1	0.508409	Exact	#NUM!	Exact	#VALUE!	Exact
1E+05	1E+05	-1	0.500841	Exact	#NUM!	0.679499	#VALUE!	Exact
1E+07	1E+07	-1	0.500084	Exact	#NUM!	0.992000	#VALUE!	Exact
1E+09	1E+09	-1	0.500008	Exact	#NUM!	0.994979	#VALUE!	Exact

3.3. Poisson distribution

If a random variable X has a Poisson $Po(\lambda)$ distribution with the mean parameter λ , then the Excel function POISSON, with the logical argument *cumulative* = FALSE ($\Sigma = 0$), computes for a given k , the probability $Pr(X = k)$. As can be seen from Table 4, Excel 97, 2000, and XP can correctly compute the extreme lower tail probabilities while giving no result for central and upper tail probabilities. Microsoft "fixed" this problem by replacing a poor algorithm with another poor algorithm, so that Excel 2003 and 2007 can compute the correct results in the central part of this distribution while rounding extreme tail probabilities down to zero. This performance is significantly inferior in comparison to those of Gnumeric and Calc, which can provide exact results even for probabilities as small as 1.73×10^{-286} . In addition, Poisson's cumulative distribution function is known to converge toward 0.5 for large values of $k = \lambda$, however, results computed with Excel 2003 and 2007 seem to converge toward 1, resulting in answers with zero accurate digits. In this case, Calc also fails to calculate the correct answer, however, by returning an error message it avoids misleading the user. Gnumeric can provide the exact result even for $\lambda = 1 \times 10^9$.

3.4. Gamma distribution

If a random variable X has a gamma $\Gamma(\alpha, \beta)$ distribution with the shape parameter α and the scale parameter β , then the Excel function GAMMADIST, with the logical argument *cumulative* = TRUE ($\Sigma = 1$), computes, for a given x , the probability $Pr(X \leq x)$. Excel versions 97 through 2007 print out these probabilities with 6 or more significant digits by default while only 5 of these digits may be correct as can be seen from Table 5. Also, for $x \leq 0.11$, Excel may return the error #NUM!, which is a fault known since Excel 2000,⁶ and still not fixed in Excel 2007. Our tests show that Gnumeric and Calc seem to provide exact results with at least six accurate digits in this distribution.

3.5. Inverse standard normal distribution

If a random variable X has a standard normal $N(\mu, \sigma^2)$ distribution with the mean $\mu = 0$ and variance $\sigma^2 = 1$, then the Excel function NORMSINV computes, for a given probability p , the value of k such that $Pr(X \leq k) = p$. As Table 6 shows,

⁶ See Microsoft Knowledgebase Article #215214.

Table 5

Gamma distribution with parameters ($x, \alpha, \beta = 1, \Sigma = 1$)

		EXACT	ELV Ed.2	EXCEL 97/2K/XP/2003/2007	CALC 2.3.0	GNUMERIC 1.7.11
0.1	0.1	0.827552	Exact	#NUM!	Exact	Exact
0.2	0.1	0.879420	Exact	0.879419	Exact	Exact
0.2	0.2	0.764435	Exact	0.764434	Exact	Exact
0.2	0.2	0.816527	Exact	Exact	Exact	Exact
0.3	0.3	0.726957	Exact	Exact	Exact	Exact
0.4	0.3	0.776381	Exact	0.776380	Exact	Exact
0.4	0.4	0.701441	Exact	Exact	Exact	Exact
0.5	0.4	0.748019	Exact	0.748018	Exact	Exact
0.5	0.5	0.682689	Exact	Exact	Exact	Exact
0.6	0.5	0.726678	Exact	Exact	Exact	Exact

Table 6

Inverse standard normal distribution with parameter (p)

		EXACT	ELV Ed.2	EXCEL 97/2K	EXCEL XP	EXCEL 2003/2007	CALC 2.3.0	GNUMERIC 1.7.11
5E-1	0	Exact	Exact	Exact	5.47142E-10	-1.39214E-16	Exact	Exact
1E-1	-1.28155	Exact	Exact	Exact	Exact	Exact	Exact	Exact
1E-2	-2.32635	Exact	Exact	-2.32634	Exact	Exact	Exact	Exact
1E-3	-3.09023	Exact	Exact	-3.09024	-3.09025	Exact	Exact	Exact
1E-4	-3.71902	Exact	Exact	-3.71947	-3.71909	Exact	Exact	Exact
1E-5	-4.26483	Exact	Exact	-4.26546	-4.26504	Exact	Exact	Exact
1E-6	-4.75342	Exact	Exact	-4.76837	-4.75367	Exact	Exact	Exact
1E-7	-5.19934	Exact	Exact	-5.000000	-5.19969	Exact	Exact	Exact
1E-15	-7.94135	Exact	Exact	-5.000000	-7.93587	Exact	Exact	Exact
1E-16	-8.22208	Exact	Exact	-5.000000	-8.29366	Exact	Exact	Exact
1E-100	-21.2735	Exact	Exact	-5.000000	-8.29366	Exact	Exact	Exact
1E-197	-29.9763	No solution	No solution	-5.000000	-8.29366	Exact	Exact	Exact
1E-198	-30.0529	No solution	No solution	-5.000000	-8.29366	-30	Exact	Exact
1E-300	-37.0471	No solution	No solution	-5.000000	-8.29366	-30	Exact	Exact

this function in Excel 97 and 2000 is unacceptably bad and can return results with no accurate digits. Microsoft "fixed" this problem twice⁷ by first tightening the convergence criteria for the iterative inversion algorithm in Excel XP and then by improving the accuracy of the normal distribution in Excel 2003. Except for the boundary case $p = 0.5$, this Excel function can now be judged acceptable by the standards of Knüsel for probabilities larger than 1×10^{-200} . On the other hand, as Knüsel (2005) mentions, it is easier to compute small probabilities due to the faster convergence in the tails of a distribution and there is no reason why a good algorithm, such as the one used in Calc or Gnumeric, cannot return exact results with six accurate digits for $p > 1 \times 10^{-300}$ for this procedure. Finally, the reason ELV cannot return a solution for the last three examples is a design decision by Knüsel (2003) to not support probabilities less than 1×10^{-100} .

3.6. Inverse chi-square distribution

If a random variable X has a chi-square $\chi^2(n)$ distribution with n degrees of freedom, then the Excel function CHINV computes, for a given probability p , the value of k such that $Pr(X > k) = p$. Excel 97, 2000, and XP report these quantiles with 9 or 10 significant digits, however, as can be seen from Table 7, only one or two of these digits may be correct if p is small. The algorithm used in Excel 2003 and 2007 seems to be acceptable by the standards of Knüsel (1995), however, it also shows instability when p is small or n is large. This function in Calc is unacceptably bad and can return misleading output with zero accurate digits. Our tests show that it is very difficult to fault Gnumeric in this distribution as well.

3.7. Inverse beta distribution

If a random variable X has a beta $Be(\alpha, \beta)$ distribution with the non-negative shape parameters α and β , then the Excel function BETAINV computes, for a given probability p , the value of k such that $Pr(X \leq k) = p$. As Table 8 shows, this function in Excel versions 97 through 2007 is very unreliable and can return incorrect results for not-so-small probability values. The corresponding BETAINV function in Calc also produces inaccurate output for small p values and needs to be fixed immediately. Gnumeric, on the other hand, can return exact results even when p is extremely small.

⁷ See Microsoft Knowledgebase Article #826772.

Table 7
Inverse chi-square distribution with parameters (p, n)

p	n	EXACT	ELV Ed2	EXCEL 97/2K/XP	EXCEL 2003/2007	CALC 2.3.0	GNUMERIC 1.7.11
2E-1	1	1.64237	Exact	1.64238	1.64238	Exact	Exact
2E-1	5	7.28928	Exact	7.28927	Exact	Exact	Exact
1E-1	1	2.70554	Exact	2.70554	Exact	Exact	Exact
1E-1	5	9.23635	Exact	9.23635	Exact	Exact	Exact
1E-5	1	19.5114	Exact	19.5037	Exact	Exact	Exact
1E-5	5	30.8562	Exact	30.7987	Exact	Exact	Exact
1E-6	1	23.9281	Exact	24.3664	Exact	23.9293	Exact
1E-6	5	35.8882	Exact	35.6115	Exact	35.8896	Exact
1E-7	1	28.3740	Exact	#NUM!	#NUM!	28.5474	Exact
1E-7	5	40.8630	Exact	#NUM!	#NUM!	40.9382	Exact
1E-12	1	50.8441	Exact	#NUM!	#NUM!	48	Exact
1E-12	5	65.2386	Exact	#NUM!	#NUM!	320	Exact
0.48	778	779.312	Exact	#NUM!	#NUM!	Exact	Exact
0.50	780	779.333	Exact	Exact	Exact	Exact	Exact
0.52	782	779.353	Exact	#NUM!	#NUM!	Exact	Exact

Table 8
Inverse beta distribution with parameters ($p, \alpha = 5, \beta = 2$)

p	EXACT	ELV Ed2	EXCEL 97/2K/XP/2003/2007	CALC 2.3.0	GNUMERIC 1.7.11
1E-1	4.89684E-01	Exact	Exact	Exact	Exact
1E-2	2.94314E-01	Exact	2.94315E-01	Exact	Exact
1E-3	1.81386E-01	Exact	1.81396E-01	Exact	Exact
1E-4	1.12969E-01	Exact	1.13037E-01	Exact	Exact
1E-5	7.07371E-02	Exact	7.03125E-02	7.07370E-02	Exact
1E-6	4.44270E-02	Exact	4.29688E-02	4.44268E-02	Exact
1E-7	2.79523E-02	Exact	3.12500E-02	2.78232E-02	Exact
1E-8	1.76057E-02	Exact	3.12500E-02	1.65959E-02	Exact
1E-9	1.10963E-02	Exact	3.12500E-02	2.83309E-03	Exact
1E-10	6.99645E-03	Exact	3.12500E-02	2.83309E-04	Exact
1E-11	4.41255E-03	Exact	3.12500E-02	2.83309E-05	Exact
1E-12	2.78337E-03	Exact	3.12500E-02	2.83309E-06	Exact
1E-13	1.75589E-03	No solution	3.12500E-02	2.83309E-07	Exact
1E-100	6.98827E-21	No solution	3.12500E-02	0.00000E+00	Exact

Table 9
Inverse t-distribution with parameters ($p, n = 1$)

p	EXACT	ELV Ed2	EXCEL 97/2K/XP	EXCEL 2003/2007	CALC 2.3.0	GNUMERIC 1.7.11
2E-1	1.37638E+00	Exact	Exact	Exact	Exact	Exact
1E-1	3.07768E+00	Exact	Exact	Exact	Exact	Exact
1E-2	3.18205E+01	Exact	3.18210E+01	Exact	Exact	Exact
1E-3	3.18309E+02	Exact	3.18289E+02	Exact	Exact	Exact
1E-4	3.18310E+03	Exact	3.18527E+03	Exact	Exact	Exact
1E-5	3.18310E+04	Exact	3.17383E+04	Exact	Err:523	Exact
1E-6	3.18310E+05	Exact	3.12500E+05	Exact	Err:523	Exact
1E-7	3.18310E+06	Exact	5.00000E+06	Exact	Err:523	Exact
1E-8	3.18310E+07	Exact	5.00000E+06	1.00000E+07	Err:523	Exact
1E-11	3.18310E+10	Exact	5.00000E+06	1.00000E+07	Err:523	Exact
1E-12	3.18310E+11	Exact	5.00000E+06	1.00000E+07	Err:523	3.18327E+11
1E-13	3.18310E+12	No solution	5.00000E+06	1.00000E+07	Err:523	3.18338E+12
1E-100	3.18310E+99	No solution	5.00000E+06	1.00000E+07	Err:523	#NUM!

3.8. Inverse t distribution

If a random variable X has a Student's t (n) distribution with n degrees of freedom, then the Excel function TINV computes, for a given probability p , the value of k such that $\Pr(|X| > k) = p$. As Table 9 shows,⁸ Excel 97, 2000, and XP can produce output with only two or three digits if p is small. The "improved" search method used in Excel 2003 and 2007 performs

⁸ While Excel's TINV function requires a 2-tailed probability, following the earlier studies on Excel's statistical distributions, here we report the 1-tailed probabilities computed by using $2 \times p$.

Table 10
Inverse F distribution with parameters ($p, n_1 = 1, n_2 = 1$)

p	EXACT	ELV Ed2	EXCEL 97/2K/XP	EXCEL 2003/2007	CALC 2.3.0	GNUMERIC 1.7.11
5E-1	1	Exact	Exact	Exact	Exact	Exact
4E-1	1.89443E+00	Exact	Exact	Exact	Exact	Exact
2E-1	9.47214E+00	Exact	9.47216E+00	Exact	Exact	Exact
1E-1	3.98635E+01	Exact	3.98636E+01	Exact	Exact	Exact
1E-2	4.05218E+03	Exact	Exact	Exact	Exact	Exact
1E-3	4.05284E+05	Exact	4.05312E+05	Exact	Exact	Exact
1E-4	4.05285E+07	Exact	4.05275E+07	Exact	Exact	Exact
1E-5	4.05285E+09	Exact	#N/A	1.00000E+09	Err:523	Exact
1E-6	4.05285E+11	Exact	#N/A	1.00000E+09	Err:523	Exact
1E-12	4.05285E+23	Exact	#N/A	1.00000E+09	Err:523	Exact
1E-13	4.05285E+25	No solution	#N/A	1.00000E+09	Err:523	Exact
1E-100	4.05285E+199	No solution	#N/A	1.00000E+09	Err:523	Exact

better,⁹ however, instead of returning the answer "1000000000" it would be preferable if Excel returned "#NUM!" so as not to mislead the users for small p . The corresponding TINV function in Calc, while acceptable by Knusel's standards, also has room for improvement. Here, we see that Gnumeric 1.7.11 also can provide an inaccurate answer for $p < 1 \times 10^{-11}$, however, this is already fixed in Gnumeric 1.7.91 which was released a few weeks after we notified the developers about this problem.

3.9. Inverse F distribution

If a random variable X has an $F(n_1, n_2)$ distribution with n_1 and n_2 degrees of freedom, then the Excel function FINV computes, for a given probability p , the value of k such that $\Pr(X > k) = p$. As Table 10 shows, this function in Excel 97, 2000 and XP is unacceptably bad. The "improved" function in Excel 2003 and 2007 performs better but can become unstable and return "1000000000" for not-so-small probability values. Calc also fails to compute the correct answer for $p < 1 \times 10^{-4}$, however, inconvenient it may be, this is acceptable from an accuracy perspective because the user is not misled. Gnumeric, on the other hand, can provide exact results even when $p = 1 \times 10^{-100}$.

4. Conclusion

It is our understanding that the algorithms for the computation of various statistical distributions in Excel 2007 can be inaccurate and/or unstable, and therefore can be unsafe to use. In particular, for the binomial, Poisson, inverse standard normal, inverse beta, inverse student's t , and inverse F distributions, it is possible to obtain results with zero accurate digits. Our results also show that the alternative Gnumeric and OpenOffice.org Calc programs, which employ dissimilar subroutines for the computation of statistical distributions, provide better accuracy in general in comparison to Excel 2007. In particular, Gnumeric can uniformly return exact values with at least six digits of accuracy for probabilities as small as 10^{-300} in all of our tests except one, and this is already fixed within a few weeks after we contacted the developers about the problem. Calc has important numerical difficulties for the computation of the quantiles of various distributions including the inverse chi-square, inverse beta, inverse t , and inverse F distributions. Once notified about the problems, Calc developers expressed their intention to correct these flaws with the upcoming OpenOffice.org version 2.4.

A new trend in computing is Web-based applications, which facilitate the collaborative creation and modification of documents over the Internet in real time. The recently introduced Google Spreadsheet, a part of the Google Docs online service, is one such application competing with the microcomputer software evaluated in this study. Our cursory examination of Google Spreadsheet finds gross errors in the accuracy of the standard normal, binomial, hypergeometric, and Poisson distributions. Consequently, a thorough evaluation of this application is necessary to help researchers and practitioners make the decision whether to move from the PC to the grid.

It is a known fact that Excel is commonly used in a wide range of decision making processes from options trading to research in physical laboratories. Offering statistical functionality in a computer program is a serious matter and it brings important responsibilities to the software vendor. Microsoft has repeatedly shown its lack of interest to this concernment by releasing new versions of Excel without first correcting the problems documented by different authors on various different occasions. Because of Microsoft's lack of commitment to accuracy, it is now possible to find on the Internet various users' custom scripts and macros for proper computation of statistical distributions in Excel.¹⁰ It is unclear when, if at all, Microsoft will properly fix Excel's inaccurate procedures for all of which there are free, well-known, and reliable alternatives. Meanwhile, researchers should continue to avoid using the statistical functions in Excel 2007 for any scientific purpose.

⁹ It is worth noting that, for all of our tests, Excel's distribution functions corresponding to the quartile functions are able to calculate the correct probability, given the inverse. Consequently, the problems noted for various inverse distribution functions are strictly shortcomings of the "improved" numerical inversion routines used in Excel 2003 and 2007 rather than side effects of poor distribution routines.

¹⁰ One such example is "Smith's Workbook" (available at <http://members.aol.com/ianjdsmith/iansNApage.htm>), which contains a set of VBA procedures for calculating probability related quantities in Excel with significantly improved accuracy.

Acknowledgements

I would like to express my appreciation to the anonymous referees, whose comments led to significant improvements to this manuscript.

References

- de Laat, P., 2005. Copyright or copyleft? An analysis of property regimes for software development. *Research Policy* 34, 1511–1532.
- Keeling, K.B., Pavur, R.J., 2007. A comparative study of the reliability of nine statistical software packages. *Computational Statistics and Data Analysis* 51, 3811–3831.
- Knüsel, L., 1995. On the accuracy of the statistical distributions in GAUSS. *Computational Statistics and Data Analysis* 20, 699–702.
- Knüsel, L., 1998. On the accuracy of statistical distributions in Microsoft Excel 97. *Computational Statistics and Data Analysis* 26, 375–377.
- Knüsel, L., 2002. On the reliability of Microsoft Excel XP for statistical purposes. *Computational Statistics and Data Analysis* 39, 109–110.
- Knüsel, L., 2003. Computation of Statistical Distributions - Documentation of Program ELV. <http://www.stat.uni-muenchen.de/~knuesel/>, [Online; retrieved July 4, 2007].
- Knüsel, L., 2005. On the accuracy of statistical distributions in Microsoft Excel 2003. *Computational Statistics and Data Analysis* 48, 445–449.
- McCullough, B.D., 2000. The accuracy of Mathematica 4 as a statistical package. *Computational Statistics* 15, 279–299.
- McCullough, B.D., 2001. Does microsoft fix errors in Excel? In: *Proceedings of the 2001 Joint Statistical Meetings*. American Statistical Association, Alexandria, VA, [CD-ROM].
- McCullough, B.D., 2004. Fixing Statistical Errors in Spreadsheet Software: The Cases of Gnumeric and Excel. http://www.csdassn.org/software_reports/gnumeric.pdf, [Online; retrieved July 4, 2007].
- McCullough, B.D., Vinod, H.D., 1999. The numerical reliability of econometric software. *Journal of Economic Literature* 37, 633–655.
- McCullough, B.D., Wilson, B., 1999. On the accuracy of statistical procedures in Microsoft EXCEL 97. *Computational Statistics and Data Analysis* 31, 27–37.
- McCullough, B.D., Wilson, B., 2002. On the accuracy of statistical procedures in Microsoft Excel 2000 and Excel XP. *Computational Statistics and Data Analysis* 40, 713–721.
- McCullough, B.D., Wilson, B., 2005. On the accuracy of statistical procedures in Microsoft Excel 2003. *Computational Statistics and Data Analysis* 49, 1244–1252.
- Yalta, A.T., 2007. The numerical reliability of GAUSS 8.0. *The American Statistician* 61, 262–268.



Contents lists available at ScienceDirect
Computational Statistics and Data Analysis

journal homepage: www.elsevier.com/locate/csda



Microsoft Excel's 'Not The Wichmann–Hill' random number generators

B.D. McCullough

LeBow College of Business, Department of Decision Sciences, Drexel University, 19104 Philadelphia, PA, United States

ARTICLE INFO

Article history:
Available online 13 March 2008

ABSTRACT

Microsoft attempted to implement the Wichmann–Hill RNG in Excel 2003 and failed; it did not just produce numbers between zero and unity, it would also produce negative numbers. Microsoft issued a patch that allegedly fixed the problem so that the patched Excel 2003 and Excel 2007 now implement the Wichmann–Hill RNG, as least according to Microsoft. We show that whatever RNG it is that Microsoft has implemented in these versions of Excel, it is not the Wichmann–Hill RNG. Microsoft has now failed twice to implement the dozen lines of code that define the Wichmann–Hill RNG.

© 2008 Elsevier B.V. All rights reserved.

1. Introduction

The desiderata for a (pseudo-)random number generator (RNG) are simple (Ripley, 1990):

1. produce numbers that are approximately uniform;
2. produce numbers that are approximately independent in a moderate number of dimensions;
3. have a sufficiently long period; and
4. be reproducible from a simply-specified starting point but, unless a starting point is specified, should be unpredictable.

Point (1) has to do with the quality of the random numbers (RNs). If the RNs are not uniform, then a simulation that requires uniform input will not yield the correct conclusion. For example, a simple univariate Monte Carlo integration on the interval $[0, 1]$ might produce an incorrect answer if the RNG's output, that is used as input to the integration process, is not uniform on $[0, 1]$. Point (2) also has to do with the quality of the RNs. Truly random numbers are uncorrelated across dimensions; if pseudo-random numbers are correlated across dimensions then an incorrect answer can be produced. For example, if a particular RNG produces uniform RNs but these RNs are correlated in two dimensions, it might produce an accurate answer for a univariate Monte Carlo integration, but an inaccurate answer for a bivariate Monte Carlo integration. To determine whether an RNG satisfies Points (1) and (2), theoretical and empirical considerations must be entertained. See L'Ecuyer (1994, Section 2.2) for a discussion of theoretical properties of an RNG. Empirical properties are assessed using a collection of randomness tests. A decade ago, the primary collection of tests was Marsaglia's (1996) DIEHARD suite of randomness tests. However, it was later supplanted by a much better program, TESTU01 (L'Ecuyer and Simard, 2007), which was first released in 2002. TESTU01 offers three batteries of tests. If an RNG passes the "Small Crush" battery (which takes a few minutes of CPU time) then apply the "Crush" battery (which can take a few hours) and if it passes all those, apply "Big Crush" (which can take an entire day). The DIEHARD tests, in contrast, take only several seconds to run on a modern PC. Since its release, TESTU01 has been the preferred program for empirical testing of RNGs since it can uncover flaws in RNGs that pass other suites of tests (Gentle, 2003, p. 85). See McCullough (2006) for a review of TESTU01.

Point (3) is important because Points (1) and (2) are generally valid only for a fraction of the period length, so if one wants to generate a million RNs, an RNG with a period of several million should be used. L'Ecuyer (1992, p. 306) warned that "No generator should be used for any serious purpose if its period (or, at least, a lower bound of it) is unknown." Over ten years

E-mail address: bdmccullough@drexel.edu.

0167-9473/\$ – see front matter © 2008 Elsevier B.V. All rights reserved.
doi:10.1016/j.csda.2008.03.006

Table 1
First five values from VBA RNG

VBA RN	VBA RN
0.705547511577606	11837123
0.533424019813537	8949372
0.579518616199493	2727092
0.289562463760376	4828052
0.301948010921478	965847

ago the minimum accepted period length for an RNG was 2^{60} (L'Ecuyer, 1994). For example, Ripley (1990) suggests that the period (p) should be an order of magnitude larger than the number of calls (n) to the RNG, $p \gg 200n^2$. Finally, Point (4) is important for purposes of debugging and testing, as well as reproducibility. This is typically effected by allowing the user to "set the seed" of the RNG, so that every time the same seed is set, the result is that the same sequence of RNs is produced by the RNG.

McCullough and Wilson (1999) criticized Excel 97, as well as Excel 2000 and Excel XP (McCullough and Wilson, 2002), for failing to meet any of these standards. Microsoft, with Excel 2003, attempted to remedy 1–3 by implementing the Wichmann and Hill (1982) (WH) RNG, whose 37 lines of code (23 of which are comments) are presented in the Appendix to this paper.¹ As noted by McCullough and Wilson (2005), Microsoft failed to implement correctly the WH RNG: instead of producing numbers between zero and unity, Excel would eventually produce negative numbers. Since the WH RNG cannot produce negative numbers, it was fair to conclude that whatever RNG Microsoft implemented in Excel 2003, it was not the WH RNG. Microsoft never disclosed the actual algorithm that was implemented or the reason that its implementation produced negative numbers (a seeming impossibility if one examines the WH code). We shall refer to this unknown RNG as "Microsoft Excel's Not The Wichmann–Hill - 1" (MENTWH1) since its true algorithm and other properties are unknown. With respect to MENTWH1, Microsoft could not achieve the desiderata for an RNG.

Microsoft soon issued a patch for Excel (see Microsoft Knowledge Base Article 834520), claiming that the patched version of Excel 2003 implements the WH RNG; further, this same RNG exists in Excel 2007 (see Microsoft Knowledge Base Article 828795). We show that this RNG also is not the WH RNG – Microsoft has failed again to implement correctly the dozen lines of code that constitute the WH RNG. We shall refer to this new Excel RNG as MENTWH2. In Section 2 we show one way to verify that an RNG has been correctly implemented. In Section 3 we show how this method can be applied to the WH RNG. In Section 4 we show the results of applying this method to the WH and other RNGs, including the MENTWH2 RNG in Excel 2007. Section 5 offers the conclusions.

2. How to check an RNG

Given a pseudo-random number generator whose period is less than about 2^{60} and the values produced in the range 0.0 to 1.0 are given in double precision, then it may well be possible to predict the sequence from just a few values. For a simple example, consider the RNG in Microsoft Visual Basic which is not a good RNG. It fails all the tests in Small Crush, and L'Ecuyer and Simard (2007, p. 30) refer to it as a "toy generator". However, its ubiquity makes it useful for illustrative purposes. It is given by the relationship:

$$X_{n+1} = (1140671485 \times X_n + 12820163) \bmod 2^{24}. \quad (1)$$

The value of the random number (in the range 0–1) is given by $X/2^{24}$; hence we can recover the value of X_n from the floating point value given. This depends upon the floating point values having at least 24 bits of accuracy.

Since VBA always uses the same initial value (i.e., seed) $X_0 = 327680$, it always produces the same string of numbers. The first five numbers produced by this algorithm are presented in Table 1.

To verify that the algorithm in (1) is used to produce these numbers, we apply the following steps:

1. multiply the first uniform RN, 0.705547511577606, by the modulus (2^{24}) to obtain 11837123
2. substitute 11837123 for X_n in (1) to obtain 8949372
3. divide 8949372 by 2^{24} to obtain the second RN, 0.533424139.

Note that this answer agrees with the VBA answer to only six digits. One possible reason is that our calculations have been in double precision while VBA RNs apparently are produced via single precision.

This procedure for determining whether a sequence of RNs is generated by (1) will work for any two successive RNs produced by the RNG given by (1). The reader can easily verify that it applies to the 999th and 1000th RNs from the VBA RNG, which are 0.311508715152740 and 0.467859745025635, respectively. However, when this method is applied to the WH RNG, even double precision will not be sufficient to achieve more than a couple of digits of accuracy, and so specialized computation methods will be necessary.

3. The Wichmann–Hill RNG

Wichmann and Hill (1982) published a random number generator with an approximate period of 2^{43} that became very popular. It uses a combination of three linear congruential RNGs to obtain a uniform random number W . First, the "seeds" I_x , I_y and I_z are initialized to take on integer values between 1 and 30,000. Then the following equations are executed to advance the seeds and compute W :

```

 $I_x := 171 \times (I_x \bmod 177) - 2 \times (I_x \div 177);$ 
 $I_y := 172 \times (I_y \bmod 176) - 35 \times (I_y \div 176);$ 
 $I_z := 170 \times (I_z \bmod 178) - 63 \times (I_z \div 178);$ 
if  $I_x < 0$  then
   $I_x := I_x + 30269;$ 
if  $I_y < 0$  then
   $I_y := I_y + 30307;$ 
if  $I_z < 0$  then
   $I_z := I_z + 30323;$ 
 $W := I_x/30269.0 + I_y/30307.0 + I_z/30323.0;$ 
return  $W - \lfloor W \rfloor;$ 

```

If users could set the seeds, it would be an easy matter to compute successive values of the WH RNG and thus ascertain whether Excel is correctly generating WH RNGs. We pointedly note that Microsoft programmers obviously have the ability to set the seeds and to verify the output from the RNG; for some reason they did not do so. Given Microsoft's previous failure to implement correctly the WH RNG, that the Microsoft programmers did not take this easy and obvious opportunity to check their code for the patch is absolutely astounding. Since Microsoft does not enable users to set the seeds, computing successive values is problematic as it requires inverting the above equations; that is, finding I_x , I_y and I_z from the values of W . What makes this problem difficult is the rounding error introduced in the floating point calculation of W . Fortunately, reverse-engineering the RNG is not necessary to determine whether a sequence of RNs has been produced by the WH RNG.

Zeisel (1986) has shown that the WH RNG can be written as a simple linear congruence generator in the form:

$$X_{n+1} = 16555425264690 \times X_n \bmod 27817185604309 \quad (2)$$

which we will call the Zeisel recursion. An implementation of the generator would not actually compute the advancing of the seed using this formula because it would overflow on most standard computers, but this does not matter to us, as the above equation is more than suitable for present purposes. The "seed" X_n can be recovered from the random value by multiplying the random value by the modulus, 27817185604309. Then the Zeisel recursion can be used to compute X_{n+1} which, divided by the modulus, yields the next (tentative) random value. This tentative next random value can be compared to the actual next random value and, if they are the same, it can be safely concluded that the RNs are produced by the WH RNG. If the next tentative random value and the next actual random value differ, then it can be concluded that the RNs are not produced by the WH RNG.

The difficulty with using the Zeisel recursion to compute random values is that it requires long integer calculations. However, these can easily be undertaken using a simple Python program since the Python language handles large integers. The slight complication in this is that the first guess at the "seed" will be an approximation to the exact value due to the rounding of the floating point operations. We can allow for this by trying a few integer values either side of the initial guess that is formed by the product of the random value and the modulus.

4. Results

To test the efficacy of the program, in a single-blind experiment, the author sent seven strings of RNs to a colleague, who was to use the program and report back whether the RNs were from a WH RNG or not. The set of strings included three WH RNG strings (one from R and two hand-coded in different languages) and four other RNGs. In all cases, the program correctly identified each string of RNs as being from a WH RNG or some other RNG. To demonstrate the program's methodology, Table 2 includes three sets of RNs, a Wichmann–Hill from R, Marsaglia's Super-Duper from R, and the Microsoft Excel 2007 RNG. We are, quite sure that the RNs from "R" are double precision. Microsoft claims to perform calculations in double precision (Microsoft Knowledge Base Article 78113), and we will take Microsoft at its word, though Kahan (2006) has some interesting observations on this point.

The interested reader can verify our RNs from R version 2.4.0, for which we used the following code:

```

set.seed(123, kind="Wichmann--Hill")
#set.seed(123, kind="Super-Duper")
x <- runif(5)
format(as.data.frame(x), digits=15)

```

¹ This RNG is not to be confused with the modern, long-period version of the WH RNG described in Wichmann and Hill (2006).

Table 2

Random numbers from various generators

Row	WH from R	Super-Duper from R	MENTWH2 from R
1	0.4462944498771981	0.5741969865221056	0.602128391977151
2	0.8905663455043975	0.09802140463961234	0.07449083663303
3	0.145225216378	0.77073464165276	0.319135092083749
4	0.340739650891	0.746977628673834	0.256824220935559
5	0.703185593285	0.583258033586610	0.65949763269207

As Microsoft does not permit a seed to be set for the Excel RNG, there can be no audit trail for Excel RNs, and the reader will just have to trust us that the numbers we present come from Excel 2007.

4.1. WH RNG from R

As a preliminary, we observe that using double precision on a standard PC does not provide sufficient accuracy. First we attempt to recover the seed by multiplying an RN by the modulus,

$$0.4462944498771981 \times 27817185604309 = 12414655546407.$$

Maybe this tentative seed is corrupted by rounding error; if so we will have to try nearby values (in fact this is the correct seed). Next we use this value as X_n in the Zeisel recursion and form the product $16555425264690 \times 12414655546407$ which cannot be accurately computed using standard methods on a PC. Standard methods can only handle sixteen digits before resorting to scientific notation, and so for this product we get 2.055299020854103E26 (instead of the correct answer 205529902085410284553468830). Consequently, applying the mod operation to this incorrect product will not produce a very accurate answer, but let us do so:

$$2.055299020854103E26 \bmod 27817185604309 = 24780451348480$$

whereas the correct answer is:

$$205529902085410284553468830 \bmod 27817185604309 = 24773049325847.$$

We divide the incorrect result by the modulus in an attempt to obtain the next RN in the sequence, $24780451348480/27817185604309 = 0.890832440814624$ which is accurate only to three digits (as opposed to the correct answer, which is $24773049325847/27817185604309 = 0.8905663455043975$). Three digits of accuracy is insufficient for present purposes. Python can effect all the calculations quite accurately, and we should be able to reproduce several digits of a WH RN. Below is some output from running the program. It first calculates a tentative seed of 12414655546406, rejects it, and then tries ten values on either side of the first guess (we show only four on either side), eventually settling on 12414655546407:

```
seed 12414655546402 fail -4
seed 12414655546403 fail -3
seed 12414655546404 fail -2
seed 12414655546405 fail -1
seed 12414655546406 fail 0
seed 12414655546407 pass 1
seed 12414655546408 fail 2
seed 12414655546409 fail 3
seed 12414655546410 fail 4
```

Having identified the correct seed, the program then computes the Zeisel recursion a few times:

```
Starting seed value: 12414655546407
0.890566345504
0.145225216378
0.340739650891
0.703185593285
0.261453152604
```

Comparing these numbers to the first column of Table 2 shows, to no great surprise, that the implementation of the WH RNG in R is correct.

4.2. Super-duper RNG from R

We undertake exactly the same steps as in the previous subsection. First we attempt to recover the seed by multiplying an RN by the modulus, $0.5741969865221056 \times 27817185604309 = 15972544147520.3$ which we round to 15972544147520. We tried 10 values either side as the seed (though we show results for only four), but each of these possible seeds failed to be consistent with the Zeisel recursion.

```
seed 15972544147516 fail -4
seed 15972544147517 fail -3
seed 15972544147518 fail -2
seed 15972544147519 fail -1
seed 15972544147520 fail 0
seed 15972544147521 fail 1
seed 15972544147522 fail 2
seed 15972544147523 fail 3
seed 15972544147524 fail 4
```

This is to be expected because these Super-Duper RNs are not generated by the Zeisel recursion. Hence we conclude, to no great surprise, that Marsaglia's Super-Duper RNG does not produce WH RNs.

4.3. MENTWH2 RNG

We repeat the same steps as before, multiplying the first random value by the modulus: $0.602128391977151 \times 27817185604309 = 16749517237252.53$ which we round to 16749517237252. Checking ten values (though we show only four) on either side of this produces no seed that is consistent with the Zeisel recursion.

```
seed 16749517237248 fail -4
seed 16749517237249 fail -3
seed 16749517237250 fail -2
seed 16749517237251 fail -1
seed 16749517237252 fail 0
seed 16749517237253 fail 1
seed 16749517237254 fail 2
seed 16749517237255 fail 3
seed 16749517237256 fail 4
```

Thus, to great surprise we show that Excel's the so-called WH RNG does *not* produce WH RNs; to wit: Microsoft has again failed to implement correctly the WH RNG. Hence we call it not the WH RNG, but MENTWH2 RNG.

A referee pointed out that perhaps our inability to locate the starting point for the generator was due to problems with the rounding in Excel. Kahan (2006) has documented some rounding problems, but in our case, no actual calculation is being performed in Excel, just the output of the random values. To be sure that the rounding to 15 digits has not caused any problem, the program was re-run by trying over 1000 possible values for a match, without success.

5. Conclusions

Twice Microsoft has attempted to implement the dozen lines of code that define the Wichmann and Hill (1982) RNG, and twice Microsoft has failed, apparently not using standard methods for verifying that an RNG has been correctly implemented. Consequently, users of Excel's "rand" function have been using random numbers from an unknown and undocumented RNG of unknown period that is not known to pass any standard tests of randomness. Given Microsoft's first failure to implement the WH RNG correctly, Microsoft should have taken special care to ensure that it was done correctly the second time. Microsoft did not do this. The second failure demonstrates clearly that the first failure (and the second, as well) was not some sort of "unfortunate accident" but, rather, a systemic failure on Microsoft's part.

Of course, one has to wonder why Microsoft chose an antiquated RNG instead of a modern RNG. In the first place, its period is only 2^{43} which, following Ripley's (1990) suggestion, can only support a couple of hundred thousand calls to the RNG. Further, Microsoft appealed to Marsaglia's (1996) DIEHARD suite of randomness tests (see Microsoft Knowledge Base Article 828795) instead of the much more stringent tests offered by L'Ecuyer and Simard's (2007) TESTU01 which has been available since 2002. Perhaps TESTU01 was released too late for Microsoft to make changes in Excel 2003, but certainly it was released in time for Microsoft to make changes in Excel 2007.

Modern empirical testing of RNGs requires that the RNG be programmed into the testing software; it is not practical to make a file of RNs and read them into the software. Since Microsoft has not made available the algorithm for MENTWH2, we cannot subject it to empirical testing. Hence, whether MENTWH2 passes any modern battery of randomness tests is unknown.

It will be interesting to see what Microsoft chooses to do. Will Excel users have to wait until Excel 2010 to get good random numbers, or will Microsoft offer a patch for Excel 2007? Will Microsoft once again attempt to implement the Wichmann–Hill RNG and, if so, will the third time be a charm? Or will Microsoft offer an RNG more suitable to the 21st century and, if so, will Microsoft program it correctly? Whatever Microsoft chooses to do, it should ensure that users have the ability to verify the RNG, since Microsoft has demonstrated that users cannot rely on Microsoft's claims to have done it correctly.

In the meantime, Excel users will have to content themselves with an unknown RNG of unknown period that is not known to pass any standard battery of tests for randomness. At least now, users will not mistakenly believe that they are using WH RNGs. Finally, Microsoft should do Messrs. Wichmann and Hill the courtesy of taking their names off the RNG in Excel.

Acknowledgements

Thanks to two referees for the extremely useful reports, to Ian Smith and Jerry Lewis for comments, and to various colleagues for assistance.

Appendix. The Wichmann–Hill code

```

real function random()
c
c   Algorithm AS 183 Appl. Statist. (1982) vol.31, no.2
c
c   Returns a pseudo-random number rectangularly
c   distributed between 0 and 1.
c
c   IX, IY and IZ should be set to integer values
c   between 1 and 30000 before the first entry.
c
c   Integer arithmetic up to 30323 is required.
c
c   integer ix, iy, iz
c   common /randc/ ix, iy, iz
c
c   ix = 171 * mod(ix, 177) - 2 * (ix / 177)
c   iy = 172 * mod(iy, 176) - 35 * (iy / 176)
c   iz = 170 * mod(iz, 178) - 63 * (iz / 178)
c
c   if (ix .lt. 0) ix = ix + 30269
c   if (iy .lt. 0) iy = iy + 30307
c   if (iz .lt. 0) iz = iz + 30323
c
c   If integer arithmetic up to 5212632 is available, the
c   the preceding 6 statements may be replaced by:
c
c   ix = mod(171 * ix, 30269)
c   iy = mod(172 * iy, 30307)
c   iz = mod(170 * iz, 30323)
c
c   on some machines, this may slightly increase
c   the speed. the results will be identical.
c   random = amod(float(ix) / 30269.0 + float(iy) / 30307.0 +
c   *          float(iz) / 30323.0, 1.0)
c   return
c   end

```

References

- Gentle, J.E. 2003. Random Number Generation and Monte Carlo Methods, 2nd ed. Springer-Verlag, New York.
- Kahan, W., 2006. How futile are mindless assessments of roundoff in floating-point computation? <http://www.cs.berkeley.edu/~wkahan/Mindless.pdf>.
- L'Ecuyer, P., 1992. Testing random number generators. In: Swain, J.J., Crain, R.C., Wilson, J.R. (Eds.), Proceedings of the 1992 Winter Simulation Conference. IEEE Press, NY, pp. 305–313.
- L'Ecuyer, P., 1994. Uniform random number generation. Annals of Operations Research 53, 77–120.
- L'Ecuyer, P., Simard, R., 2007. TESTU01: A C library for empirical testing of random number generators. ACM TOMS 33 (4), Article 22.
- Marsaglia, G., 1996. DIEHARD: A battery of tests of randomness. <http://stat.fsu.edu/pub/diehard>.

- McCullough, B.D., 2006. A review of TESTU01. Journal of Applied Econometrics 21 (5), 677–682.
- McCullough, B.D., Wilson, B., 1999. On the accuracy of statistical procedures in microsoft excel 97. Computational Statistics & Data Analysis 31 (1), 27–37.
- McCullough, B.D., Wilson, B., 2002. On the accuracy of statistical procedures in microsoft excel 97. Computational Statistics & Data Analysis 40 (4), 713–721.
- McCullough, B.D., Wilson, B., 2005. On the accuracy of statistical procedures in microsoft excel 2003. Computational Statistics & Data Analysis 49 (4), 1244–1252.
- Ripley, B.D., 1990. Thoughts on pseudorandom number generators. Journal of Computational and Applied Mathematics 31, 153–163.
- Wichmann, B.A., Hill, I.D., 1982. Algorithm AS 183: An efficient and portable pseudo-random number generator. Applied Statistics 31 (2), 188–190.
- Reprinted, with a correction, in Hill I.D., Griffiths P. (Eds.), Applied Statistics Algorithms, Ellis Horwood, Chichester, 1985.
- Wichmann, B.A., Hill, I.D., 2006. Generating good pseudo-random numbers. Computational Statistics & Data Analysis 51 (3), 1614–1622.
- Zeisel, H., 1986. Remark on algorithm AS 183. Applied Statistics 35, 89.



Contents lists available at ScienceDirect
Computational Statistics and Data Analysis
 journal homepage: www.elsevier.com/locate/csda



It's easy to produce chartjunk using Microsoft® Excel 2007 but hard to make good graphs

Yu-Sung Su

Department of Political Science, The Graduate Center, The City University of New York, 365 Fifth Avenue, New York, NY 10016, USA

ARTICLE INFO

Article history:
 Available online 13 March 2008

ABSTRACT

The purpose of default settings in a graphic tool is to make it easy to produce good graphics that accord with the principles of statistical graphics, e.g., [Tuft, E.R., 1990. *Envisioning Information*. Graphics Press, Cheshire, Conn. Tuft, E.R., 1997. *Visual Explanations: Images and Quantities, Evidence and Narrative*, 2nd Edition. Graphics Press, Cheshire, Conn. Cleveland, W.S., 1993. *Visualizing Data*. Hobart Press, NJ. Cleveland, W.S., 1994. *The Elements of Graphing Data*, rev. edition. AT&T Bell Laboratories, Murray Hill, NJ. Wainer, H., 1997. *Visual revelations: Graphical tales of fate and deception from Napoleon to Ross Perot*. Copernicus, New York. Spence, R., 2001. *Information Visualization*. ACM Press & AddisonWesley, New York. and Few, S., 2004. *Show Me the Numbers*. Analytic Press, Hillsdale, NJ]. If the defaults do not embody these principles, then the only way to produce good graphics is to be sufficiently familiar with the principles of statistical graphics. This paper shows that Excel graphics defaults do not embody the appropriate principles. Users who want to use Excel are advised to know the principles of good graphics well enough so that they can choose the appropriate options to override the defaults. Microsoft® should overhaul the Excel graphics engine so that its defaults embody the principles of statistical graphics and make it easy for non-experts to produce good graphs.

Published by Elsevier B.V.

"Data graphics should draw the viewer's attention to the sense and substance of the data, not to something else. The data graphical form should present the quantitative contents. Occasionally artfulness of design makes a graphic worthy of the Museum of Modern Art, but essentially statistical graphics are instruments to help people reason about quantitative information" (Tuft, 1997, p. 91).

1. Introduction

A well-constructed chart can bring the data to life, revealing the substantive information. A well-designed graphic tool makes it easy for users to achieve this goal. Microsoft® Excel performs poorly in this respect. On the one hand, statisticians seldom use Excel because it is not a sophisticated statistical package, unlike other alternatives (e.g., SAS®, SPSS®, S-PLUS®, STATA®, and R), it does not have programmable procedures to do statistical analysis and make graphs. On the other hand, common users who just want to glimpse and/or pre-process data or make some simple graphs are attracted by the WYSIWYG (the acronym of What You See Is What You Get) nature of Excel. Users can easily modify the design of a chart by clicking buttons. They do not have to type or remember codes or commands for making graphs. Nonetheless, convenience should not supersede the accuracy and clarity of demonstrating the data through graphs. In particular, the default chart types in Excel often distort graphics with redundant symbols, fill-ins and other extraneous graphical elements. Edward Tuft describes these kinds of redundant "interior decoration of graphics" as *chartjunk* (1997, p. 83) because they do not tell viewers anything new about the data.

E-mail address: ys463@columbia.edu.

0167-9473/\$ – see front matter. Published by Elsevier B.V.
 doi:10.1016/j.csda.2008.03.007

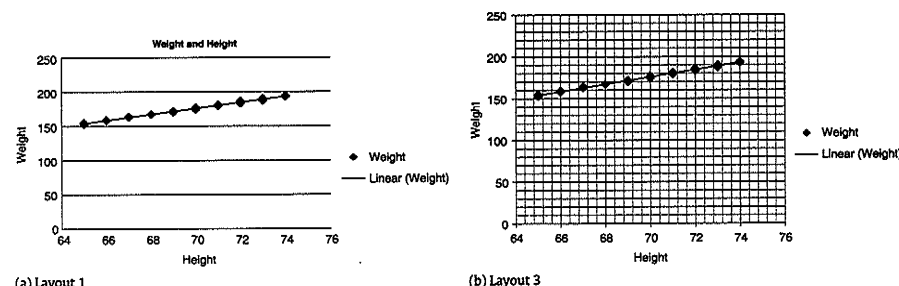


Fig. 1. Scatter chart.

Microsoft® has just released Excel 2007 and claims that the new graphics engine has enhanced Excel's graphical capabilities. Various reviews of Excel 2007 have shown that, the novel graphical interface notwithstanding, Excel 2007 continues to provide users with the clutter and fluff in its default chart types (Few, 2006; Walkenbach, 2007). This paper evaluates Excel's graphical capabilities on its default chart types and, since there is no obvious improvement, the following evaluation is applicable to the previous Excel products (Excel 95, Excel 2000, Excel XP, and Excel 2003).¹ This paper is not a manual about how to make good statistical graphics in Excel (See O'Day (2007), Peltier (2007) and Vidmar (2007) and various blog entries in Juice Analytic, <http://www.juiceanalytics.com>), nevertheless, it employs the philosophy of what counts as good graphics as discussed in works by Tuft (1990, 1997), Cleveland (1993, 1994) and Wainer (1997) when assessing various Excel charts.

2. Defaults matter!

Defaults matter because most users rely heavily on the default settings of the software they use. Nowadays, computing power and various graphical software have advanced to a level where everyone can be an artist, creating various graphics by him/herself. However, greater power comes with greater responsibility. Excel should guide its users toward good graphical design and not toward making junk charts. Microsoft should be more careful about the default settings on Excel charts.

Fig. 1 plots the linear relationship between height and weight of 40 people using the default scatter chart with two different default Excel layouts. These plots remind us how crude an Excel chart can look. Here, and also in the plots of the following sections, I try to adjust the Excel default settings as little as possible so that I can convey the idea of how nonsensical the default graphical outputs look. Some compromises and adjustments had to be made though, but *chartjunk* remain.

This type of chart is useful when we want to display a relationship between two variables. Fig. 1 shows that weight and height are positively related. The overlying regression line depicts this linearity. These charts are poorly designed for several reasons:

- The over-stretched axes hinder viewers' ability to read the data. The exaggerated y-axis simply distorts the data revelation. Sometimes we do want the scale to go all the way to the zero point because zero has its meaning in certain cases. This is not the case for this example. No one is expected to have zero weight, thus it is legitimate to start the y-axis from the minimum weight. Rescaling the y-axis in this way will help viewers identify more data points from the chart. In the current design, users can only observe 10 points where in fact there are 40.
- The redundant grid lines and the excess axis labels create nothing but clutter that messes up the graphical area. The purpose of the grid lines is supposed to help viewers better identify the absolute position of the data points, especially when they are dispersed over the plotting rectangle. Because this is not the case here, the grid lines are distracting. In particular, Fig. 1(b) overly uses grid lines, which obscure the crux of this chart—the data. Likewise, both the y-axis and x-axis have too many labels. This is messy.
- The legend and labels are not only superfluous; they are also wrong. The dots here certainly do not only mean weight. Rather, they represent 40 different people with the relative coordination of their weight and height. Also, the line indicator here is redundant. A simple graph like this does not need a legend and labels. Even when labeling is necessary, a better strategy is to use direct labeling as opposed to a legend.
- The filled diamond data points fail to reveal the overlain data points. A better choice would be to use open circles.

¹ In contrast to previous Excel products, Excel 2007 has reduced the number of its default chart types from fourteen to seven; the new categories are: Column, Line, Pie, Bar, Area, Scatter, and other charts.

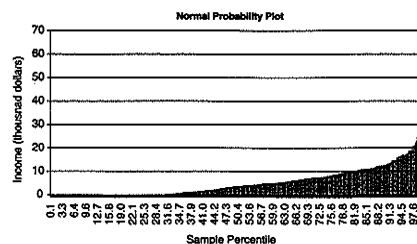


Fig. 2. Normal probability plot.

- The colour scheme can be too light for viewers to read and is also not suitable for publication. In particular, two axis lines are light grey in colour. This implies that the two axes are graphical elements equal in importance to the major grid lines, which is a wrong premise. Similarly, using light blue colour for the data dots is not illuminating enough in print.
- The most untenable default option is that there are no tick marks in Fig. 1(a)! In some default layouts, Excel removes tick marks from the chart. Viewers are going to have a hard time reading Excel graphics without the help of tick marks because Excel tends to mark too many labels on axes. Use of both tick marks and axis labels is a better way to limit the excess use of axis labels.²

Fig. 2 is another poorly designed chart in Excel 2007. This type of chart is called a normal probability plot (Chambers et al., 1983), which is a graphical technique for visually evaluating whether or not a data set is normally distributed. The vertical axis is the ordered response value whereas the horizontal axis is normal order statistic median. Thus, if the data set is normally distributed, we would expect to see a straight line.³ Excel 2007 keeps the option of making this plot in the "data analysis tool pack" add-in. The plot was a line chart in the previous versions of Excel. Excel 2007 makes it into a column chart. This is a wrong choice because the length of each bar does not mean anything and it might mislead the readers to think there is an accumulative pattern for the data. Nevertheless, the excess axis label is an untenable design.

In short, Excel's default chart types are problematic in that they embed superfluous graphical elements and design poorly. Unfortunately, Excel has more than just these defects. The following sections show the way in which Excel misleads users who will end up wasting their time creating fancy designs which do not help to make their data stand out. Instead, they create junk charts.

3. Create junk charts in Excel: Just by clicks of buttons!

In this section, I pretend to be a typical user who tries to make charts using Excel 2007. I select five charts to show the way in which the default charts in Excel can violate the principle of statistical graphics.⁴ One common mistake of these charts is the use of three-dimensional (3-D) plotting. 3-D objects often overwhelm the comprehensibility of human eye. Ironically, they can attract users' as well as readers' attention easily. Nonetheless, this does not justify the use of 3-D plotting. In addition, 3-D plotting in Excel often conveys null information and always distorts the data. As shown in Fig. 3, when users scroll down a chart menu, the options immediately next to the 2-D chart types are a variety of 3-D chart types. Excel leads users to produce bad graphical designs just by clicks of buttons.

3.1. Line chart

A line chart is useful to spot trends of continuous data. Sometimes, a line chart is a good choice when you want to compare multiple data series on the same baseline. Fig. 4 shows the monthly expense of six different items from January to June, plotted as a 3-D line chart. This type of chart obscures the data display. Viewers cannot make the comparison of multiple data series because of the overlapping lines. The additional axis, which labels the data series, fails to identify the six data series clearly. Nevertheless, the width of each line is *chartjunk*, which shows null information.

² See Cleveland (1994) for the general principles of using tick marks and axis labels, and Wilkinson (2005) for getting The optimal number of axis labels and tick marks by rescaling the data.

³ The normal probability plot in Excel is incorrectly programmed: it shows a straight line for uniform data, not for normal data (See McCullough and Heiser, 2008, Section 5).

⁴ I use R (R Development Core Team, 2007) to create fake data to make the following charts.

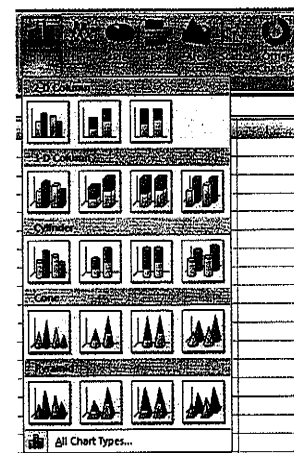


Fig. 3. The chart menu in Excel 2007.

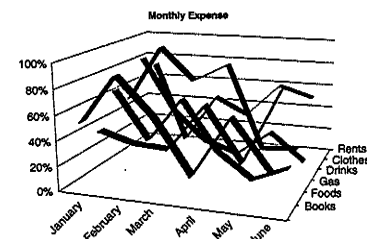


Fig. 4. Line chart.

3.2. Pie chart

A pie chart is useful when users want to show relative proportions of a whole. However, a pie chart is almost never effective.⁵ Tufte (1997) wrote, "the only worse design than a pie chart is several of them" (p. 178). The fact that a pie chart can only convey a single series of the data means that a dot plot is an excellent alternative because it can show data positions along a common scale rather than rely on pie chart angles. Fig. 5 compares the proportion of aid money among five major donor countries, plotted as a 3-D pie chart. The 3-D effect and the angle of this pie chart distort the data. Viewers would misidentify Japan (22%) over the US (33%) as the largest aid money donor. Another reason not to use a pie chart is that it wastes data ink. A pie chart uses color, size, angle, 3-D effect, and fill-in to convey only one single message from the data—the relative proportion to a whole. In this case, a table can do a better job than a chart even though graphs are usually better than tables in summarizing numerical information of the data. Don't use graphics to "merely decorate a few numbers" (Tufte, 1997, p. 83).

3.3. Area chart

An area chart is a special case of a line chart in which the area below the line has been coloured. An area chart is not an example of an effective chart, either. The comparison between data series is often misleading and obscured due to the size

⁵ There is a lively debate about the cons and pros of the pie chart as a means to display data (Cleveland and McGill, 1984; Simkin and Hastie, 1987; Spence, 1990; Carswell et al., 1991; Rangecroft, 2003). Some argue that the pie chart is still useful in showing a good qualitative view of the data, particularly when there are not too many small pieces. For more recent discussion with this regard, please refer to Few (2007) and a blog entry by Zach Gernigani, "The problem with pie charts" and the comments posted to this entry, <http://www.juiceanalytics.com/writing/2006/12/the-problem-with-pie-charts>.



Fig. 5. Pie chart.

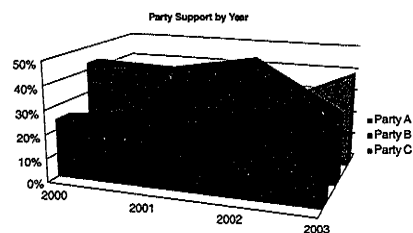


Fig. 6. Area chart.

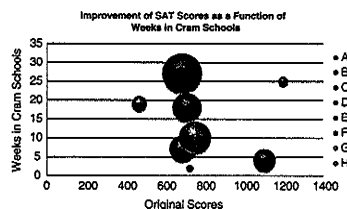


Fig. 7. Bubble chart.

of the areas. Fig. 6 shows the change of support rates for three parties from 2000 to 2003, plotted as a 3-D area chart. The use of the third dimension helps viewers to better identify three different parties, though at the expense of obscuring the data. In particular, the area of Party C is veiled by the area of Party B. Excel offers the solutions such as rotating the charts or using transparency. But these solutions are not going to work in print.

3.4. Bubble chart

A bubble chart is a special case of scatter chart. It displays an additional data series by altering the size of the bubbles. However, a bubble chart sometimes can be misleading because people perceive the mapping from value to the size of the bubbles in a way that is not easy to quantify. Fig. 7 shows the improvement in SAT scores for eight students as the function of weeks cramming in school, plotted as a 3-D bubble chart. The bubble chart shows the relative improvement in SAT scores effectively by the size of the bubbles. However, an open circle is a better choice than a colour-filled bubble because data can be masked due to overlaps. The legend on the right of the chart is redundant. A better legend would indicate what the size of the bubbles means. Nevertheless, the colour fill-in and the 3-D effect create a false impression in which these extra dimensions convey no information.

3.5. Bar chart

A bar chart displays each data point as a horizontal bar, the length of which corresponds to the value. A 90 degrees counterclockwise bar chart is a column chart. A bar chart is a superior choice over a column chart when you have a lengthy category label. Fig. 8 displays the change of monthly support rate for three candidates, plotted as a 3-D stacked pyramid chart, a subtype of the bar chart. The stacked pyramids fail to clearly show the relative proportion of the three candidates.

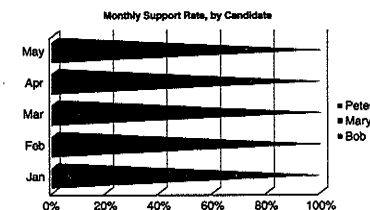


Fig. 8. Pyramid chart.

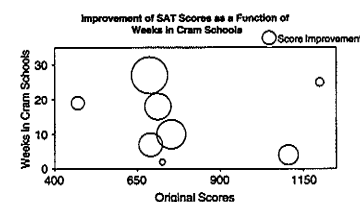


Fig. 9. Improved bubble chart.

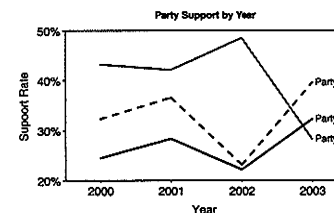


Fig. 10. Improved area chart using line chart.

In addition, the pyramid shape suggests that there are differences between the candidates in some perspective other than support rates. In fact, shape here is just *chartjunk*. It contains no information but distorts the data.

4. Effective charting in Excel is laborious!

It is disappointing that Excel 2007 did not provide any new chart types or make any significant improvement. In order to make charts that effectively display the data, users have to expend great effort to clean up the *chartjunk* in Excel defaults. Here are some steps to do this:

- No fill-in or unnecessary colouring and shadow.
- 3-D effect is seldom a good option to use.
- No busy grid lines and over-crowded labels on axes. Adjust scale for both axes and avoid overstretching axes.
- Tick marks are still necessary, especially when you have data that is dispersed over the plotting rectangle. Tick marks help viewers to better locate data in the relative coordinates.
- Direct labeling is a better alternative to identify data objects than a legend.

Fig. 9 is a revised version of Fig. 7 after cleaning up *chartjunk*. The chart effectively reveals the overlapped data points with open circle. The improved legend shows that the circle size means the improvement in SAT scores. The succinct labeling on both axes enables viewers to concentrate on reading the data information.

Fig. 10 is an improved version of Fig. 6. A line chart is more effective than an area chart in comparing different trend lines. Fig. 10 clearly demonstrates that 2002 and 2003 are the two years that mark distinctive changes among the three trend lines. The pattern is veiled in Fig. 6 because of *chartjunk*.

Fig. 11 is another example of effective charting that summarizes the relationships between an outcome variable and predictors. The plot shows the regression coefficients of various predictors on income, plotted as a stock chart.

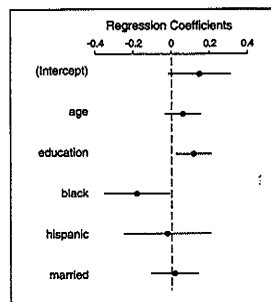


Fig. 11. Stock chart.

Dots represent regression estimates. Bars show the 95% confidence intervals of the estimates (or ± 1.96 standard errors). If the intervals of an estimate cross the zero reference line, the estimate is statistically insignificant. Viewers do not have to count stars in a regression table. This figure demonstrates the way in which an effective chart can help viewers to digest numerical information.

To make junk charts in Excel, users just need a few clicks of the buttons. To clean up *chartjunk*, they require more than a few. Some Excel experts have written and developed Excel macro add-ins to do the clean-up jobs (See O'Day (2007), Peltier (2007) and Vidmar (2007)). However, users could save time and energy by searching for and installing these macros. They could have focused more on their works had they had an excellent chart tool at the beginning. A good graphic design certainly helps effectively bring data to life. Excel just does not offer a good guidance, instead it gives users many other chart styles that fail to communicate.

5. Conclusions

A properly-designed chart can help people get the most information out of data and an excellent graphic tool helps people to achieve this task without much labour. The purpose of default settings in a graphic tool is to make it easy to produce good graphics that accord with the principles of statistical graphics. If the defaults do not embody these principles, then the only way to produce good graphics is to be sufficiently familiar with the principles of statistical graphics (Tuft, 1990, 1997; Cleveland, 1993, 1994; Wainer, 1997; Spence, 2001; Few, 2004).

This paper has shown that the default chart types in Excel 2007 do not embody these appropriate principles. Instead, these charts create *chartjunk* that hinder peoples' ability to comprehend the data. Some users have developed add-ins and instructions to redress the malfunctions of the Excel default chart settings and thereby enable users to produce better graphics (O'Day, 2007; Peltier, 2007; Vidmar, 2007). Nevertheless, those who want to use Excel are advised to get to know the principles of good graphing well enough so that they know how to choose the appropriate options to override the defaults. Microsoft® should overhaul its graphics engine so that it embodies the principles of statistical graphics and makes it easy for non-experts to produce good graphs.

Acknowledgements

I would like to thank Dr. Andrew Gelman for his advices on the earlier draft and the suggestion for the title. I would also like to thank Stephen Few for his comments and Masanao Yajima for his inputs on this paper. Finally, I would like to thank three anonymous referees for their valuable comments and suggestions on how to improve this paper.

References

- Carswell, C.M., Frankenberger, S., Bernhard, D., 1991. Graphing in depth: Perspectives on the use of three-dimensional graphs to represent lower-dimensional data. *Behaviour & Information Technology* 10 (6), 459–474.
- Chambers, J.M., Cleveland, W.S., Kleiner, B., Tukey, P.A., 1983. *Graphical Methods for Data Analysis*. PWS Publishers: Duxbury Press, Boston.
- Cleveland, W.S., 1993. *Visualizing Data*. Hobart Press, N.J.
- Cleveland, W.S., 1994. *The Elements of Graphing Data*, rev. edition. AT&T Bell Laboratories, Murray Hill, N.J.
- Cleveland, W.S., McGill, R., 1984. Graphical perception: Theory, experimentation, and application to the development of graphical methods. *Journal of American Statistical Association* 79, 531–554.
- Few, S., 2004. *Show Me the Numbers*. Analytic Press, Hillsdale, NJ.
- Few, S., May 2006. Excel's new charting engine: preview of an opportunity missed. http://www.perceptualedge.com/articles/b-eye/excels_new_charting_engine.pdf.
- Few, S., August 2007. Perceptual Edge, Save the pies for dessert. http://www.perceptualedge.com/articles/visual_business_intelligence/save_the_pies_for_dessert.pdf.

- McCullough, B.D., Heiser, D.A., 2008. On the accuracy of statistical procedures in microsoft excel 2007. *Computational Statistics & Data Analysis* 52 (10), 4570–4578.
- O'Day, K.D., 2007. Data analysis and visualization with excel tools and charts. <http://processtrends.com/>.
- Peltier, J., 2007. Jon's excel and charting pages. <http://peltiertech.com/>.
- R Development Core Team, 2007. R: a language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria.
- Rangecroft, M., 2003. As easy as pie. *Behaviour & Information Technology* 22 (6), 421–426.
- Simkin, D., Hastie, R., 1987. An information-processing analysis of graph perception. *Journal of the American Statistical Association* 82 (398), 454–465.
- Spence, I., 1990. Visual psychophysics of simple graphical elements. *Journal of Experimental Psychology: Human Perception and Performance* 16, 683–692.
- Spence, R., 2001. *Information Visualization*. ACM Press & Addison Wesley, New York.
- Tufts, E.R., 1990. *Envisioning Information*. Graphics Press, Cheshire, Conn.
- Tufts, E.R., 1997. *Visual Explanations: Images and Quantities, Evidence and Narrative*, 2nd edition. Graphics Press, Cheshire, Conn.
- Vidmar, G., 2007. Statistically sound distribution plots in excel. *Metodološki Zvezki* 83–98.
- Wainer, H., 1997. *Visual revelations: Graphical tales of fate and deception from Napoleon to Ross Perot*. Copernicus, New York.
- Walkenbach, J., 2007. *Excel 2007 Bible*. Wiley, Indianapolis, IN.
- Wilkinson, L., 2005. *The Grammar of Graphics*, 2nd Edition. Springer, New York.



Teaching statistics with Excel 2007 and other spreadsheets

John C. Nash*

Telfer School of Management, University of Ottawa, Ottawa, Ontario K1N 9B5, Canada

ARTICLE INFO

Article history:
Available online 12 March 2008

ABSTRACT

This article considers which activities in teaching statistics may be suitable candidates for the application of spreadsheets, and whether spreadsheets in general and Excel 2007 in particular are suitable for these tasks.

© 2008 Elsevier B.V. All rights reserved.

1. A menu

This article considers the tools we use to help teach statistics, especially spreadsheets. Among spreadsheet processors, the most popular is Microsoft Excel, of which a new flavor has recently been released, which we will call Excel 2007.

In order to organize the discussion below, I will try to address the following subjects:

- What activities are involved in teaching where a spreadsheet might be used?
- Are spreadsheets in general appropriate or inappropriate to these tasks, or could they be made so?
- Where a spreadsheet might be appropriate, is Excel 2007 a good choice?

In approaching these questions, I will try to separate those matters I consider fundamental from those that are preferences, and attempt to provide good justifications for such distinctions. However, it is worth noting that style and tradition often trump substance and evidence, so the former may ultimately be the main driver of adoption of a tool, even the wrong tool (Bacon, 2006; Lyon, 2002). Bloodletting endured as a medical treatment for a very long time. Unfortunately, the debate on whether spreadsheets are suitable for use in teaching statistics has also endured, and this paper will in many places repeat arguments made before (see, for example, the content and bibliography of Dell'Omodarme and Valle (2006), as well as material referenced in the next section).

2. Context

I have been interested in spreadsheets since they were introduced in the early 1980s and have taught statistics in various venues since 1969. Nash (2006) provides references to most of my earlier work on spreadsheets and statistics, including the proof-of-concept TellTable project (www.telltale.com). I have done some work to provide test spreadsheets for Gnumeric (Nash and Goldberg, 2005). Being involved in studying open source knowledge generation models, and favoring platform independent tools, I do not regularly use Microsoft Office.

Since this paper is one of a set (see McCullough (2008), McCullough and Heiser (2008), Su (2008) and Yalta (2008)) some topics are given a very light treatment here.

3. What activities are involved in teaching where a spreadsheet might be used?

Let us first eliminate the obvious negatives, and conclude that spreadsheets are not suitable for mathematical statistics. Tools such as Maple or Mathematica are much more relevant in that area. However, the vast bulk of students and the major

* Tel.: +1 613 236 6108.

E-mail address: nashjc@uottawa.ca.

burden of teaching concerns introductory one- or two- term courses of study in other programs. This implies that we are going to need to support data management, computations and graphics for fairly simple data sets, and spreadsheets have often been chosen as the tools.

Spreadsheets get used widely in the management of teaching tasks, but as that is not specific to the teaching of statistics, I will not include such uses here, even though it has been the major application of spreadsheets in my own teaching.

4. Are spreadsheets appropriate to the tasks?

4.1. General issues

The spreadsheet paradigm, that is, results change when data is changed, is an extremely useful one for quickly testing ideas and verifying possibilities. However, most spreadsheet processors do not maintain this mode of operation across all their features. A number of computational dialogs have been introduced in different spreadsheet processors (with variation by version of the processor) where the output remains static even when the input data is changed. In Excel 2007, the Data Analysis/Regression dialog is one such tool. Using some data gathered in a teaching exercise for length and width of students' middle fingers I ran a simple regression of length versus width, then changed one length from 95 mm to 495 mm. The regression output cells did not change. However, when I drew an XY graph of the data and added a trend line, then made the same change, the trend line did change. (Thanks to a referee for pointing this out.) Here we have a mixed paradigm for the same calculation. This is like having a brake lever that sometimes acts as an accelerator—pure danger. This is not a criticism of spreadsheets in general, but of unfortunate choices made in their implementation. So readers may try tests like this, my data is available at <http://macnash.admin.uottawa.ca/~nashjc/allfinger.xls>.

Data retrieved from databases may raise even more such problems due to timing issues, since active databases do get updated regularly. Here we begin to need a concurrent version control system like CVS (<http://www.nongnu.org/cvs/>) or Subversion (<http://subversion.tigris.org/>). These tools allow multiple users to simultaneously edit files, with appropriate features and safeguards to maintain the integrity of the content, allowing for reconciliation of conflicts and recovery of different versions.

4.2. Classroom teaching of introductory statistics

With variation in detail and depth, most introductory applied statistics courses cover the following topics:

- data
- univariate, bivariate and occasionally three dimensional statistical graphics
- descriptive statistics
- probability
- discrete and continuous probability distributions
- sampling distributions
- hypothesis testing for one or two samples
- interval estimation
- count data (goodness of fit)
- analysis of variance (at least one-way, sometimes 2-way)
- regression, at least simple, often multiple

We will look very briefly at the suitability of spreadsheets for each of these.

4.2.1. Data

Modern spreadsheet processors are capable of handling tables of data that are well beyond the size needed for teaching. For the most part the user simply "enters the data" and the spreadsheet processor algorithmically decides how to handle and format it. This can lead to some unfortunate errors if the data should be treated as text but is handled as numbers, or if it is meant to record a date or time. Date handling has been the subject of much discussion over the years (e.g., O'Beirne (2000)), but are a technical matter that could be resolved with appropriate standardization activity. I personally recommend dates be recorded as numbers using a format such as YYYYMMDD to ensure proper sort order, that is, later dates sort as higher numbers. This avoids relying on software-defined date handling. While really an admission of defeat, it bypasses concerns of "locale" where we need to worry about the month/day/year versus day/month/year forms and so forth. Furthermore, there is no need to explain mathematical and software details that are beyond the scope of most undergraduates. Unfortunately, software is sometimes very fascist in "helping you" by automatically formatting cells. For example typing the characters 2007/11/17 in my Excel 2007 results in a cell with the entry 17/11/2007. If you have your regional setting with English (USA) rather than English (Canada), you will get 11/17/2007. This is why it is advisable to leave out the slashes.

A statistical issue, which again could be resolved, is that spreadsheets do not handle missing values very well (Kirschenbaum, 2006). Worse, some users hide data by various tricks such as tiny fonts or setting the same foreground and background colors for cells. This can be used, for example, to avoid displaying intermediate or verification calculations

to leave a more esthetically pleasing screen. Even without trickery, users may put spaces or other whitespace characters into cells that can then be acted upon in ways that are not equivalent to an undefined cell, or may hide data accidentally when entering colors or font sizes.

4.2.2. Univariate, bivariate and occasionally three dimensional statistical graphics

I have heard my students saying that they like “Excel” (usually synonymous with “spreadsheet” to them—an idea that suggests a student survey assignment whose results could interest trademark lawyers) for the graphics. However, I would contend that spreadsheets do not do a particularly good job compared to statistical software. Consider the ease of creating a histogram in either Minitab (www.minitab.com) or Rcmdr (<http://socserv.socsci.mcmaster.ca/~jfox/Misc/Rcmdr>) with Excel or Gnumeric. Boxplots are highly useful as summaries of distributions and particularly for comparisons, but are not available in Excel 2007, though they are a feature in Gnumeric. In Gnumeric, I found that I seem to need to group the data myself in order to get multiple boxplots; this is not very helpful when I want to do a quick analysis.

4.2.3. Descriptive statistics

Spreadsheets seem to carry out descriptive statistics calculations satisfactorily. There are also functions listed for correlation and covariance. We may want to ask what they do with missing values (see above). When there are missing values and more than two variables, there are sensible choices to only include observations that are complete or to compute pairwise correlations using all observations that have both variables recorded. I hope to build some test spreadsheets to uncover the behavior of different spreadsheet processors, and welcome assistance in this and similar efforts.

4.2.4. Probability

Spreadsheets are useful tools for dealing with most of the elementary probability topics taught in a typical statistics course.

4.2.5. Discrete and continuous probability distributions

For teaching purposes, the probability functions of spreadsheets are one of their most useful features. Moreover, unlike a calculator, we can review and edit the expressions, alter them, copy them, and so forth. There are two fringe issues. First, the spreadsheet processor may lack particular functions, though this ought to disappear in the next few years. For example, Excel lacks the cumulative hypergeometric probability; Gnumeric has a boolean switch to offer individual or cumulative hypergeometric probabilities. Second, and for the present review secondarily, the computed values may be incorrect in some way. For example, the normal distribution is computed as a body rather than tail area, so applications requiring tail areas may be problematic. The body area suffices for teaching in almost all cases, but may be insufficient for some specialized work. Try computing $(\text{normsinv}(\text{normdist}(z)) - z)$ for values of z from 8 to 8.3. While “better” functions may not always be necessary, we do need better documentation and didactic material to inform our use of software such as a spreadsheet processor.

4.2.6. Sampling distributions

Spreadsheets do quite well in illustrating sampling distributions. For Excel 2003, McCullough and Wilson (2005) had serious concerns regarding the quality of random number generators, but these are unlikely to be paramount in a teaching situation, though there is a danger in introducing students to a less-than-adequate tool for real-life work. Moreover, most statistical packages are more suitable here.

4.2.7. Hypothesis testing for one or two samples

Spreadsheets offer some tools for hypothesis testing, but in my view they are so obviously “patched” onto the application as to be less than ideal for teaching. I much prefer the statistical packages and how they present the possibilities. My principal dissatisfaction is that the user must construct the whole test – the hypothesis, the computation of the test statistic and the setup of the call to the distribution function – and then call such a function to get the equivalent of the “table value” one used to look up in the back of the textbook. Contrast this with the interface and output provided by Minitab or Rcmdr. For example, using only a few mouse clicks, the latter produces

```
> t.test (Davis$ weight, alternative = 'greater', mu = 63, conf.level = .95)
One Sample t-test
data: Davis$ weight
t = 2.6232, df = 199, p-value = 0.004692
alternative hypothesis: true mean is greater than 63
95 percent confidence interval:
 64.03611 Inf
sample estimates:
mean of x
 65.8
```

Note that the interface generates the command from a graphical user interface, so the user can later on reuse this command in a script to save effort. Also observe that we can talk of variables and observations so that the test does not need us to specify a set of cells for the test, but instead allows the user to focus on the ensemble of data we call a ‘variable’; while spreadsheets allow named ranges, we think of variables in teaching statistics.

4.2.8. Interval estimation

Here the spreadsheets are, at present, weak. They offer a “CONFIDENCE” function that gives the margin of error for a $100(1 - \alpha)\%$ interval (with default 95%) for a single-sample mean based on the Gaussian distribution.

4.2.9. Count data (goodness of fit)

Spreadsheets supply functions for the Chi-square distribution, but as in hypothesis tests do not seem to allow for the appropriate preparation of the count data so that goodness of fit and contingency tables can be quickly tested. Given that the spreadsheet is naturally a tabular form, this should make us recognize that the interests of those who build the spreadsheet processors are not aligned with those of us who teach and practice statistics.

4.2.10. Analysis of variance

ANOVA has been part of spreadsheets for quite a while, and the tools have been evolving. Single factor, two factor, and two factor with replication ANOVA is supported in Gnumeric and Excel to an extent that is suitable for course teaching. However, the interface may not suit how we wish to present material in courses. If the partitioning of the sum of squares is important in a course, then (and I am grateful to a referee for noting this), one may wish the computational details, as shown in <http://faculty.vassar.edu/lowry/ch13pt1.html>. If data is changed, users may want to ensure that the output from the dialog has been updated.

4.2.11. Regression

Regression is supported by most spreadsheets to an extent suitable for teaching. Personally, I do not find the interface convenient. As mentioned above, changing the input data does not change the regression output.

5. Is Excel 2007 a good choice as a spreadsheet processor?

Let me state up front that I do not like the 2007 incarnation of Excel. Violating the “If it ain’t broke, don’t fix it” and the KISS principles, Excel 2007 seems little more than the overstuffed toy box of a spoiled child. More than 40 icons at the top of a screen are confusing. Worse, the clutter makes it very difficult to find the few things one does need.

Do we need this clutter? Most of us use only a few features and we likely work most efficiently if we can effectively place, size and alter them. Surely this is the essence of personalization. As the menus are stored in XML, there is a possibility of altering them, but this does not seem to be easy as yet. My suspicion is that a lively third-party market for menu templates and tools will develop, even to return Microsoft Office menus to their classic appearance (e.g., for Access, there is <http://software.techrepublic.com/download.aspx?docid=301725>). When I searched “Help” using “customize menu” I was offered a party planner, a weekly meal planner, and a dinner party journal. This is worse than unhelpful.

Turning to statistics teaching, among the plethora of eye-candy graphs, I do not find histograms or boxplots, though histograms are in the Data Analysis add-in. Among the “new” graphical tools, I was told by Aaron Erickson of the possibility of colored bars scaled to the sizes of the numbers in the cells. These are activated by “Conditional formatting” in Excel 2007. Unfortunately, there is an example from Computerworld where these seem to be scaled improperly (http://www.computerworld.com/html/office_2007b/p11.html). In a sense, such bars provide a sort of histogram, so if properly implemented this could be a useful idea.

Most of the statistical tools in Excel are performed in the Data Analysis Add-in. Under Excel 2007, one needs to find this, yet the Add-ins tab at the top of the screen only shows the default add-ins. The Data Analysis Add-in is installed to the “Data” tab. Also one needs to go to the unnamed “Office” button and choose “Excel options” to activate this add-in. This is three places to look.

Turning to the add-in itself, it is worth mentioning that “histogram” is a very strange function for statisticians in that it is essentially a ‘tally’ or frequency of observations in ranges defined by the user as bin boundaries. There is no graph, and the binning is manual. Hardly a great statistical tool.

When one does finally find a tool, there is the concern about whether it does its job properly. McCullough (2004) argues that the sluggish response of Microsoft in fixing errors compromises the software. As McCullough shows, the Gnumeric programmers (who are but a few part-timers) were able to fix in several weeks the same errors that Microsoft was unable and/or unwilling to fix in several years. Excel’s statistical capabilities have been the subject of many complaints about errors and inaccuracies over many years. Finding and properly documenting such errors or weaknesses in scientific and statistical software is a difficult and tedious job, and reporting on them takes many pages of highly technical detail. Our personal and professional gratitude is owed to those who carry out such studies, such as those reported by McCullough (2004), McCullough and Wilson (2005) and McCullough (2008) and the apparently ongoing and large study by David Heiser

(<http://www.daheiser.info/>). From the point of view of accuracy, my opinion is that Excel 2007 provides sufficient accuracy for most of the tasks in elementary statistics courses, but that it is very poor pedagogy to teach students to use a tool that is inadequate for “real-life” use.

6. Conclusion

Spreadsheets are powerful tools for end-user computing (see, for instance, the program of the EuSPRIG 2004 conference at <http://eusprig.org/eusprig-2004-conference-programme.pdf>). However, the developers and vendors have vastly different agendas from statistics teachers, so we should not expect them to be well suited to our needs in the classroom. Most of us need tools that work well and offer clean, unambiguous interfaces. This means that for most spreadsheet applications I will use Gnumeric. However, for statistics, I used to use Minitab. With more students using Macintosh and Linux where a bundled textbook/Minitab package has not been available to us, I now would use the open-source R, either natively or via the Rcmdr interface.

References

- Dell'Omodarme, M., Valle, G., 2006. Teaching statistics with Excel and R. ArXiv Physics e-prints, ref. physics/0601083. <http://arxiv.org/pdf/physics/0601083.pdf>.
- Bacon, Jono, 2006. Context vs. content. <http://www.oreillynet.com/pub/wlg/9236>, Feb. 23, 2006.
- Kirschenbaum, Lief, 2006. Missing value representation in Excel. <http://tolstoy.newcastle.edu.au/R/help/06/01/18871.html>, 11 Jan. 2006.
- Lyon, Jack M., 2002. Content vs. presentation. <http://www.planetpublish.com/mainpage.asp?webpageid=231>, June 7, 2002.
- McCullough, B.D., 2004. Fixing statistical errors in spreadsheet software: The cases of Gnumeric and Excel, CSDA Statistical Software Newsletter. www.csda.org/software_reports.cfm.
- McCullough, B.D., Heiser, David A., 2008. On the accuracy of statistical procedures in Microsoft Excel 2007. Computational Statistics and Data Analysis 52 (10), 4570–4578.
- McCullough, B.D., 2008. Microsoft's 'Not the Wichmann-Hill' random number generator. Computational Statistics and Data Analysis 52 (10), 4587–4593.
- McCullough, B.D., Wilson, Berry, 2005. On the accuracy of statistical procedures in Microsoft Excel 2003. Computational Statistics and Data Analysis 49 (4), 1244–1252.
- Nash, J.C., 2006. Spreadsheets in statistical practice—another look. The American Statistician 60 (3), 287–289.
- Nash, J.C., Goldberg, Jody, 2005. Why, how and when spreadsheet tests should be used. In: Ward, David (Ed.), Proceedings of the EuSPRIG 2005 Conference: Managing Spreadsheets in the light of Sarbanes-Oxley. EuSPRIG, Greenwich UK, pp. 155–160.
- O'Beirne, Patrick, 2000. Spreadsheet year 2000 issues: Tips, traps and answers to Frequently Asked Questions (FAQ). <http://www.sysmod.com/y2ksprds.htm>.
- Su, Yu-Sung, 2008. It's easy to produce Chartjunk using Microsoft Excel 2007 but hard to make good graphs. Computational Statistics and Data Analysis 52 (10), 4594–4601.
- Yalta, A.Talha, 2008. The reliability of statistical distributions in Microsoft Excel 2007. Computational Statistics and Data Analysis 52 (10), 4579–4586.

Regular Papers